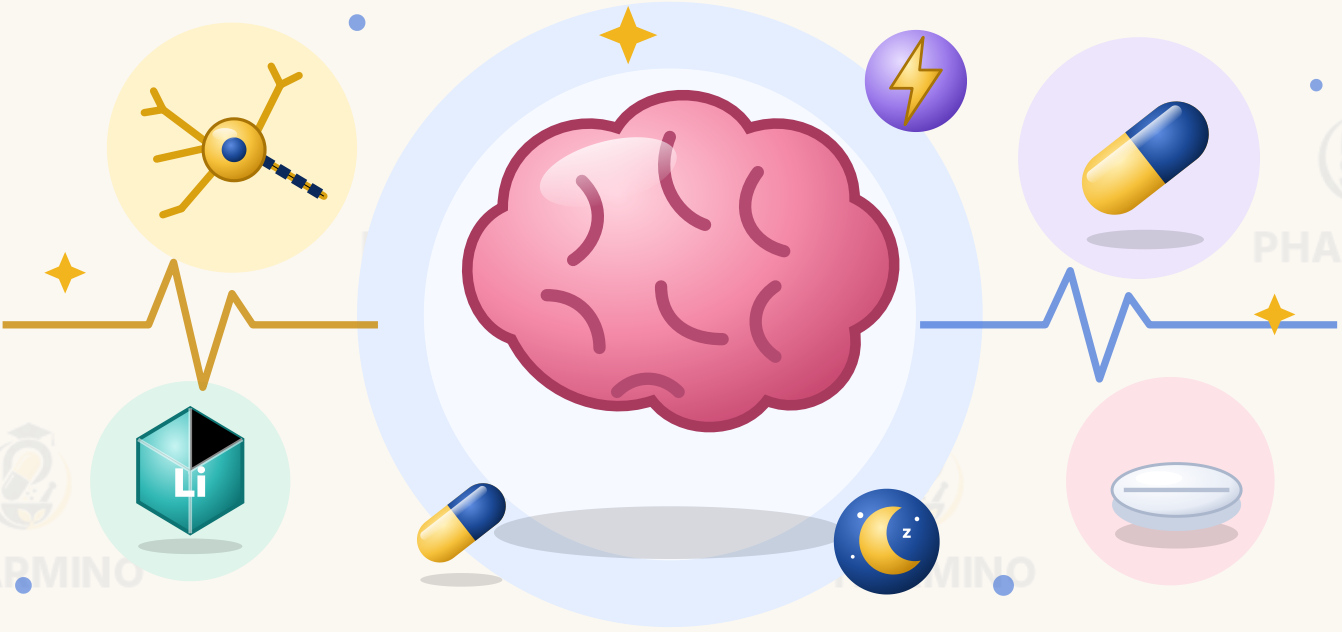
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CNS Pharmacology
Visual Review Notes

Psychiatry · Neurology · Sleep

Pearson Review Pack



Antipsychotics
Dopamine pathways · EPS · NMS

Bipolar & Lithium
Levels · toxicity · DDIs

Antidepressants
SSRI · SNRI · TCA · MAOI

Parkinson's
Levodopa · agonists · COMT

Alzheimer's
ChEIs · memantine

Epilepsy
ASMs · status epilepticus

ADHD
Stimulants · non-stimulants

Insomnia & Anxiety
Z-drugs · BZDs · buspirone

Scientifically revised edition · Oman Pearson VUE Pharmacy Licensing Exam focus · Corrections from the original notes are marked ✎ CORRECTED

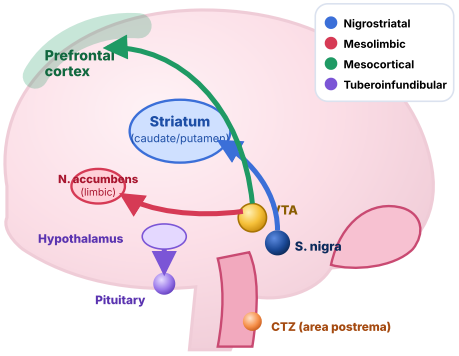
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Dopamine Pathways & Psychosis

Pearson Review Notes

1 The Four Dopaminergic Pathways (+ CTZ)

Antipsychotics block D2 receptors in ALL of them



Pathway	Normal role	Disease / D2-block effect
1. Nigrostriatal	Voluntary motor control	DA loss → **Parkinson's disease**. D2 block → **EPS** (side effect)
2. Mesolimbic	Emotion, reward, perception	**↑ DA → positive symptoms**. D2 block = **antipsychotic effect** (main target of all agents)
3. Mesocortical	Cognition, memory, motivation	**↓ DA → negative & cognitive symptoms**; D2 block may worsen them. SGAs (5-HT2A block) partly spare DA here
4. Tuberoinfundibular	DA tonically **inhibits prolactin**	D2 block → **↑ prolactin**: galactorrhoea, amenorrhoea, gynaecomastia, sexual dysfunction
CTZ (medulla)	Triggers vomiting	D2 block → **antiemetic** (prochlorperazine, droperidol, haloperidol)

✎ **CORRECTED** Original: "nausea & vomiting from mesolimbic pathway" and "antipsychotics decrease dopamine". Nausea/vomiting is mediated by the **CTZ**, not the mesolimbic pathway; antipsychotics **block D2 receptors** — they do not lower dopamine production.

2 Psychotic Disorders & Symptom Domains

Psychosis
Loss of contact with reality (hallucinations, delusions)

→

Schizophrenia
Positive + negative + cognitive symptoms ≥ 6 months

Bipolar disorder
Mania / hypomania ⇄ depression; psychosis can occur in severe mania or depression

Positive (excess) — mesolimbic ↑DA

- Hallucinations (mostly auditory)
- Delusions
- Disorganised speech / behaviour
- Respond **well** to all antipsychotics

Negative (deficit) — mesocortical ↓DA

- Flat / blunted affect
- Avolition (↓ motivation), alogia
- Anhedonia, social withdrawal
- Respond **poorly**; SGAs modestly better

Cognitive

- Impaired attention & working memory
- Poor executive function
- Major driver of disability

Receptor Blockade → Effect Map

Receptor	Result of blockade
D2	Antipsychotic, antiemetic, EPS, ↑ prolactin
5-HT2A	↓ EPS, may help negative symptoms (SGAs)
H1	Sedation, weight gain

Receptor	Result of blockade
M1	Dry mouth, constipation, urinary retention, blurred vision, confusion
α1	Orthostatic hypotension, dizziness, reflex tachycardia

High-Yield Exam Pearls

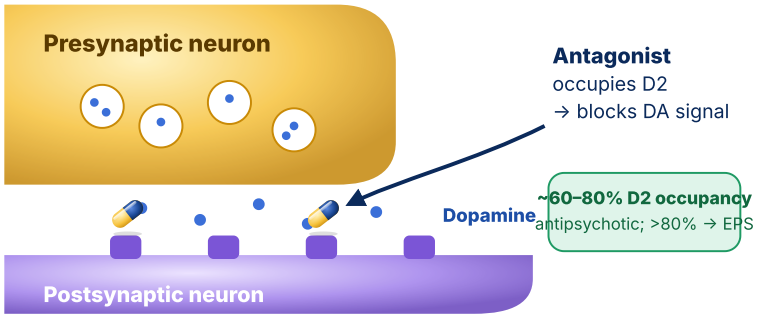
- Positive symptoms ↔ **mesolimbic ↑DA**; negative ↔ **mesocortical ↓DA**.
- EPS = nigrostriatal D2 blockade; ↑prolactin = tuberoinfundibular blockade.
- Antiemetic action of phenothiazines = **CTZ** D2 block.
- Negative symptoms are the hardest to treat.

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Schizophrenia: First-Generation Antipsychotics

Pearson Review Notes

1 Mechanism of Action — D2 Receptor Blockade



- **Typical / FGA**: potent **D2 antagonists** → treat mainly **positive** symptoms.
- Little effect on negative / cognitive symptoms.
- Higher risk of **EPS** and **hyperprolactinaemia** than most SGAs, but generally **lower metabolic risk**.
- Antipsychotic effect needs weeks; sedation and agitation control occur sooner.

2 FGA Drug Classes

Drug	Potency	Key points
PHENOTHIAZINES		
Chlorpromazine	Low	Antipsychotic > antiemetic > antihistaminic (sedative). Marked sedation, anticholinergic, α1-block (hypotension); photosensitivity, cholestatic jaundice
Thioridazine	Low	**Marked QT prolongation**; pigmentary retinopathy (dose-related) — reserved / restricted use
Trifluoperazine · Perphenazine	Medium–High	Oral maintenance agents; EPS risk
Fluphenazine	High	High EPS; **long-acting decanoate depot** injection for non-adherence
Prochlorperazine	Medium	Used mainly as an **antiemetic** / vertigo; can still cause dystonia
BUTYROPHENONES		
Haloperidol	High	Highest EPS; acute agitation/delirium; **decanoate depot**; **QT prolongation & torsades especially IV / high dose**; low metabolic risk
Droperidol	High	Antiemetic & sedation; with **fentanyl → neuroleptanalgesia**; boxed warning for QT prolongation (ECG)

First-gen potency rule

High potency haloperidol, fluphenazine → ↑ **EPS**, ↓ sedation / anticholinergic / hypotension
Low potency chlorpromazine, thioridazine → ↑ **sedation, anticholinergic, orthostasis**, ↓ EPS

Mnemonic: "High potency = High EPS; Low potency = Lots of other side effects."

High-Yield Exam Pearls

- **Haloperidol** = high potency, highest EPS, IV form → QT.
- **Thioridazine** = QT + retinopathy.
- Depots: **haloperidol & fluphenazine decanoate** (also paliperidone, aripiprazole, risperidone LAIs).
- Smoking cessation in a clozapine patient → watch for **sedation, seizures, toxicity**.

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Antipsychotics in Special Situations

Pearson Review Notes

1 Clinical Selection Scenarios

Smoking

Tobacco smoke (hydrocarbons) **induces CYP1A2** → **↓ levels** of **clozapine & olanzapine** (smokers need higher doses).
Stopping smoking → levels rise → toxicity risk; reduce dose & monitor. Nicotine replacement does not induce CYP1A2.

Diabetes / obesity

Choose **low metabolic-risk** agents: **aripiprazole, ziprasidone, lurasidone** (or high-potency FGAs e.g. haloperidol, with EPS trade-off). Avoid clozapine & olanzapine where possible.

Parkinson's + psychosis

Quetiapine (low D2 affinity) or **clozapine**; **pimavanserin** (5-HT2A inverse agonist) is licensed in some markets. **Avoid haloperidol, risperidone & other potent D2 blockers**.

Schizophrenia + mood symptoms

SGAs with mood efficacy: **olanzapine**, quetiapine, aripiprazole, risperidone; paliperidone is licensed for schizoaffective disorder.

Pregnancy

Usually **continue the antipsychotic that works** (relapse is harmful). Most data: **haloperidol, olanzapine, quetiapine**, risperidone, aripiprazole. Monitor glucose (olanzapine/quetiapine ↑ gestational diabetes risk); neonatal EPS/withdrawal possible.

Procedural sedation

Droperidol + fentanyl = neuroleptanalgesia (calm, analgesia, patient still responsive). Check ECG/QT.

✎ **CORRECTED** Original "Smoking ↑/↓ conc" (direction missing) → **smoking lowers** clozapine/olanzapine levels. "DM + schizo → haloperidol + aripiprazole" refined to **low metabolic-risk agents**; pregnancy choice is based on prior response, not a fixed 1st/2nd order.

2 Important Antipsychotic Interactions

- **Clozapine/olanzapine + smoking** (CYP1A2 induction) → ↓ levels; **fluvoxamine, ciprofloxacin** (CYP1A2 inhibitors) → ↑ levels
- Clozapine + **carbamazepine** → avoid (additive bone-marrow toxicity)
- QT drugs together (macrolides, fluoroquinolones, methadone, citalopram) → torsades
- Antipsychotics + **levodopa** → opposing effects
- CNS depressants (alcohol, BZDs, opioids) → additive sedation/respiratory depression

3 Elderly & Dementia

- **Boxed warning**: ↑ mortality (stroke, infection, cardiac) in dementia-related psychosis
- Use only for severe distress/danger, lowest dose, shortest time; review regularly
- Prefer non-drug measures; avoid strongly anticholinergic agents (Beers)
- Lewy body / Parkinson's dementia: **severe sensitivity** to D2 blockers

4 NMS vs Serotonin Syndrome vs Anticholinergic Toxicity

Feature	NMS	Serotonin syndrome	Anticholinergic toxicity
Cause	D2 antagonists (or abrupt levodopa withdrawal)	Serotonergic drugs (SSRI + MAOI, linezolid, tramadol…)	Antihistamines, TCAs, antipsychotics, atropine
Onset	Days – weeks (slow)	Hours (<24 h, rapid)	Hours
Muscles	**Lead-pipe rigidity**, bradyreflexia	**Clonus, hyperreflexia**, tremor (legs)	Normal
Skin / pupils / gut	Sweating · normal pupils · normal bowel	Sweating · **mydriasis** · **diarrhoea**, ↑ bowel sounds	**Dry, hot, red skin** · mydriasis · ↓ bowel sounds, urinary retention
Treatment	Stop drug, cooling, BZD, dantrolene, bromocriptine	Stop drugs, BZD, **cyproheptadine** (5-HT2A antagonist)	Supportive, BZD; **physostigmine** in selected cases

High-Yield Exam Pearls

- Patient with clozapine who **quits smoking** → levels ↑ → sedation, seizures.
- Parkinson's psychosis → **quetiapine / clozapine** (pimavanserin).
- Diabetes → **aripiprazole, ziprasidone, lurasidone**.

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Antipsychotic Adverse Effects

Pearson Review Notes

1 Extrapyramidal Symptoms (EPS) — Onset Timeline



EPS type	Presentation	Management
Acute dystonia	Sustained involuntary **muscle contraction**: torticollis, oculogyric crisis, trismus, laryngospasm (emergency). Risk: young males, high-potency FGA	**IM/IV anticholinergic**: benztropine or diphenhydramine (biperiden IM where available); then oral for a few days
Akathisia	Subjective inner **restlessness**, inability to sit still, pacing (not a repetitive tic)	↓ dose / switch; **propranolol (first choice)**; benzodiazepine; mirtazapine (low dose). Anticholinergics have limited benefit
Pseudo-parkinsonism	Bradykinesia, rigidity (cogwheel), resting tremor, masked face, shuffling gait	↓ dose or switch to SGA; **anticholinergic** (benztropine, trihexyphenidyl) or amantadine
Tardive dyskinesia	Involuntary choreoathetoid **orofacial** movements (lip smacking, tongue protrusion), limbs/trunk	Stop / switch (clozapine or quetiapine); **VMAT2 inhibitors: valbenazine, deutetrabenazine**. **Anticholinergics worsen TD**. Screen with AIMS

✎ **CORRECTED** Original listed **atropine** as EPS treatment and called akathisia "repetitive movement". Atropine is not used; BZDs help akathisia only; anticholinergics treat dystonia/parkinsonism and worsen TD.

2 Other Key Adverse Effects

↑ Prolactin
Galactorrhoea, amenorrhoea, gynaecomastia, ↓ libido, ↓ bone density. Highest: **risperidone, paliperidone**, high-potency FGAs. Manage: ↓ dose, switch/add **aripiprazole**; DA agonist (**cabergoline**, bromocriptine) with caution — may worsen psychosis

QT prolongation
Thioridazine, IV haloperidol, ziprasidone, droperidol, pimozide → torsades. Baseline ECG, K^+/Mg^{2+} if at risk

Orthostatic hypotension
α1-block: chlorpromazine, clozapine, quetiapine, risperidone. Titrate slowly; falls in elderly

Anticholinergic & other
Dry mouth, constipation, retention, confusion (low-potency FGAs, clozapine, olanzapine). Sedation, weight gain, ↓ seizure threshold, photosensitivity

Boxed warning: ↑ mortality in **elderly patients with dementia-related psychosis** (all antipsychotics).

3 Neuroleptic Malignant Syndrome (NMS)

rare but a medical emergency

Features
- **Hyperthermia** (high fever)
- **"Lead-pipe" rigidity**
- Altered mental status
- Autonomic instability (BP, HR, sweating)

Labs / course
- ↑ **CK**, leukocytosis
- Myoglobinuria → acute kidney injury
- Develops over days; often early in therapy or after dose ↑ / depot

Treatment
- **STOP the antipsychotic**; ICU care, cooling, IV fluids
- **Benzodiazepines**; **dantrolene** (muscle relaxant)
- **Bromocriptine** or amantadine (DA agonist)
- Re-challenge ≥ 2 weeks later with low-potency/SGA

High-Yield Exam Pearls

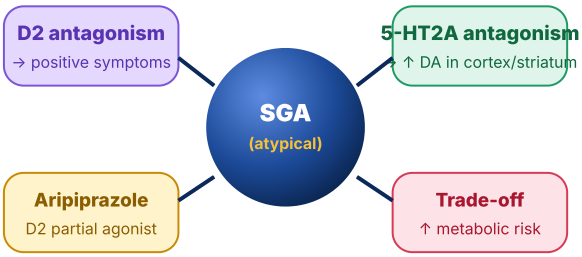
- Dystonia → **benztropine/diphenhydramine**; akathisia → **propranolol**.
- TD → **valbenazine / deutetrabenazine**; never anticholinergics.
- NMS = rigidity + fever + ↑CK → **stop drug, dantrolene, bromocriptine**.
- Clonus + diarrhoea + rapid onset = **serotonin syndrome**.

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Second-Generation (Atypical) Antipsychotics

Pearson Review Notes

1 Why SGAs Are Usually Preferred First-Line



- Block **D2 + 5-HT2A** → treat positive symptoms with **lower EPS** and less prolactin elevation (except risperidone/paliperidone).
- Modest benefit on negative symptoms; benefit for mood (bipolar, adjunct in depression).
- Overall efficacy is similar to FGAs (except **clozapine**, which is superior in treatment resistance).
- Choice is driven by **side-effect profile** and patient factors.

✎ **CORRECTED** Original: SGAs "treat both + and − symptoms and have lower side effects". More accurate: lower **EPS**, but **higher metabolic** risk; negative-symptom benefit is modest.

2 Key SGA Drugs

Clozapine
Most effective; **treatment-resistant schizophrenia** & ↓ suicidality. Severe neutropenia, myocarditis, seizures, sialorrhoea, constipation, highest metabolic risk. see next page

Olanzapine
Very effective; **marked weight gain**, dyslipidaemia, hyperglycaemia, sedation. CYP1A2 (smoking ↓ levels). IM for agitation; ODT available. Bipolar mania

Risperidone / Paliperidone
Highest prolactin elevation; dose-related EPS. Less sedating. LAIs available. Paliperidone = active metabolite (renal dosing)

Quetiapine
Low D2 affinity → **very low EPS** (Parkinson's psychosis). Sedation, orthostasis; bipolar depression & mania; metabolic risk moderate; cataract monitoring (label)

Aripiprazole
D2 partial agonist; **lowest metabolic & prolactin** risk (can ↓ prolactin). Main SE **akathisia**, insomnia; impulse-control disorders. LAI available

Ziprasidone · Lurasidone
Low metabolic risk. **Ziprasidone: QT prolongation**, take with food. **Lurasidone: take with food (≥350 kcal)**; bipolar depression; avoid strong CYP3A4 inhibitors/inducers

3 Relative Adverse-Effect Profile

qualitative ranking

Drug	Weight / metabolic	EPS	Prolactin	Sedation	QT
Clozapine	+++	0/+	0	+++	+
Olanzapine	+++	+	+	++	+
Quetiapine	++	0/+	0	+++	+
Risperidone	++	++	+++	+	+
Ziprasidone	0/+	+	+	+	++
Aripiprazole	0/+	+ (akathisia)	0 (↓)	0/+	0/+
Haloperidol (FGA)	+	+++	+++	+	++ (IV)

0 = minimal, + low, ++ moderate, +++ high. Ranking agrees with major references (e.g. Maudsley, APA guideline).

Metabolic Monitoring

- **Baseline:** weight/BMI, waist, BP, fasting glucose or HbA1c, lipids, personal/family history
- **Weight/BMI:** every visit for 6 months, then quarterly
- **Glucose & lipids:** at 12 weeks, then annually
- **Prolactin** if symptoms; **ECG** if cardiac risk/QT drugs
- Lifestyle advice; consider metformin for antipsychotic-induced weight gain

Schedule based on the ADA/APA consensus; local protocols may vary.

High-Yield Exam Pearls

- Most weight gain: **clozapine ≈ olanzapine**; least: **aripiprazole, ziprasidone, lurasidone**.
- Most prolactin: **risperidone/paliperidone**.
- Least EPS: **clozapine, quetiapine**.
- **Ziprasidone & lurasidone** must be taken **with food**.

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Clozapine & Schizophrenia Treatment Algorithm

Pearson Review Notes

1 Clozapine — Reserved for Treatment-Resistant Schizophrenia

failure of ≥ 2 adequate antipsychotic trials

ANC monitoring (label)

Start only if ANC ≥ 1500/µL
(≥ 1000/µL in benign ethnic neutropenia)

Months 0–6	Months 6–12	> 12 months
WEEKLY	every 2 weeks	MONTHLY

ANC 500–999 → interrupt therapy;
ANC < 500 → stop (severe neutropenia), haematology

✎ **CORRECTED** Original: "check CBC every 6 months". Monitoring is **weekly for the first 6 months**, then fortnightly, then monthly. The US REMS registry was removed in 2025, but **ANC monitoring is still required** by the label (risk highest in the first 18 weeks).

Adverse effect	Key points / management
Severe neutropenia (agranulocytosis)	Boxed warning; teach to report **fever, sore throat**; check ANC immediately
Myocarditis / cardiomyopathy	Mostly first **4–8 weeks**: tachycardia, chest pain, fever → troponin/CRP, ECG; stop if confirmed
Seizures	**Dose-related**; caution in epilepsy (relative, not absolute CI); valproate or levetiracetam if needed
Sialorrhoea (hypersalivation)	Especially at night: **glycopyrrolate**, sublingual atropine 1% drops, clonidine; elevate pillow
Constipation → ileus	Potentially fatal; prophylactic laxatives, monitor bowel habits
Orthostasis, syncope, sedation	Slow titration; **re-titrate from 12.5 mg** if ≥ 48 h of doses missed
Metabolic	Highest weight gain, diabetes, dyslipidaemia; fever, tachycardia, enuresis

2 Schizophrenia Pharmacotherapy Algorithm

STEP 1
First episode: **monotherapy** with an SGA (not clozapine) or FGA, chosen by side-effect profile. Start low, titrate

→

Assess at 2–6 weeks
Adequate dose & adherence? Partial / no response → **switch** to a different antipsychotic

→

STEP 2
Second monotherapy trial (SGA or FGA). Still inadequate after 2 trials = **treatment-resistant**

→

STEP 3
Clozapine (monitor ANC). Partial response → augment (e.g. aripiprazole) or **ECT**

Responder: continue maintenance — first episode at least **1–2 years**, multiple episodes often **long-term**; avoid abrupt stopping.

Non-adherence at any step → LAI: haloperidol / fluphenazine decanoate, risperidone, paliperidone, aripiprazole, olanzapine pamoate (post-injection delirium monitoring).

Persistent suicidality or aggression → consider **clozapine earlier** (only antipsychotic shown to ↓ suicide risk).

Based on APA Practice Guideline (2020/2021) and NICE CG178. Original note referred to an algorithm image — redrawn here.

Response Timeline

Days
↓ agitation, sedation, sleep improves

1–3 weeks
positive symptoms begin to improve

6+ weeks
fuller response; negative/cognitive slow & partial

High-Yield Exam Pearls

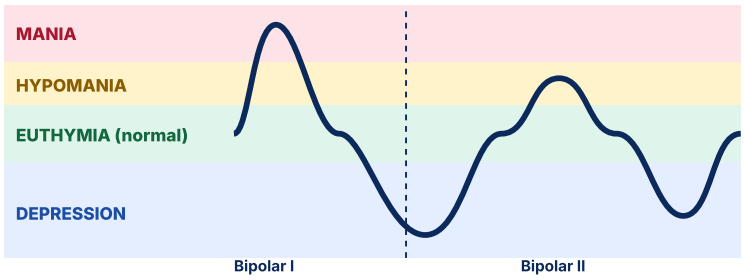
- Clozapine = **3rd-line / treatment-resistant** and suicidality.
- Clozapine fever + sore throat → **check ANC**.
- Clozapine early tachycardia + chest pain → **myocarditis**.
- Never combine clozapine with **carbamazepine**.

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Bipolar Disorder & Lithium

Pearson Review Notes

1 The Bipolar Mood Spectrum



- **Bipolar I**: ≥ 1 **manic** episode (≥ 7 days or hospitalisation; may include psychosis).
- **Bipolar II**: ≥ 1 **hypomanic** episode (≥ 4 days, no marked impairment, no psychosis) + ≥ 1 major depressive episode.
- Hypomania is less severe than mania; depressive phases dominate time spent ill.
- **Antidepressant monotherapy can trigger a switch to mania** — always combine with a mood stabiliser.

2 Lithium — Gold-Standard Mood Stabiliser

Profile

- First-line for **acute mania** and **maintenance** (prevents mania > depression)
- **↓ suicide risk** — unique benefit
- MOA not fully known: inhibits **inositol monophosphatase** & **GSK-3**
- **Narrow therapeutic index**; renal excretion only, handled like Na^+

Target serum levels

Maintenance
≈ 0.6–1.0 mEq/L (mmol/L)

Acute mania
≈ 0.8–1.2 mEq/L

Toxicity
> 1.5 mEq/L (severe > 2.5)

When to sample



- **Trough level 12 h after the last dose** (just before the next)
- Steady state ≈ **5 days** after starting or a dose change

✎ **CORRECTED** Original listed "30 min pre-dose" and "12 h post-dose" as separate times — both describe the **same 12-hour trough**. Lithium has **no antidote**; **sodium bicarbonate is the antidote for TCA** toxicity, not lithium.

3 Lithium Monitoring

Test	Baseline	Follow-up	Why
Serum lithium	—	5 days after start/dose change; then every 3 months (first year), then every 3–6 months	Narrow therapeutic index
Renal function (SCr, eGFR, urea, electrolytes/Na^+)	✔	Every 3–6 months	Nephrogenic DI, chronic kidney disease
Thyroid function (TSH)	✔	Every 6–12 months	Hypothyroidism, goitre
Calcium	✔	Every 6–12 months	Hyperparathyroidism / hypercalcaemia
ECG, weight/BMI, CBC, pregnancy test	✔ (ECG if CV risk / > 40 y)	As clinically indicated	Arrhythmia, weight gain, benign leukocytosis

Schedule summarised from NICE CG185 / BAP guidance; follow local protocol.

High-Yield Exam Pearls

- Lithium level = **12-h trough**, steady state at **~5 days**.
- Toxicity > **1.5 mEq/L**; no antidote → fluids / **haemodialysis**.
- Lithium is the only mood stabiliser with strong **anti-suicide** evidence.
- Antidepressant alone in bipolar → **mania switch**.

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Lithium Safety & Bipolar Treatment Lines

Pearson Review Notes

1 Lithium Adverse Effects & Management

GI upset
Nausea, vomiting, diarrhoea (early) → **take with food**, divide doses or use SR form

Fine tremor
Hands; worse with caffeine → **propranolol** (atenolol alternative). **Coarse tremor = toxicity**

Kidney
Polyuria, polydipsia (nephrogenic diabetes insipidus) → **amiloride**. Long-term: CKD

Thyroid / parathyroid
Hypothyroidism, goitre → **levothyroxine** (continue lithium). Hypercalcaemia

Cardiac
T-wave flattening, sinus node dysfunction, bradycardia; QT effects at toxic levels

Metabolic / skin
Weight gain, oedema; **acne, psoriasis** flare, rash, hair loss; benign leukocytosis

Pregnancy
Teratogen: first-trimester exposure → **Ebstein's anomaly** (cardiac). If continued: fetal echo, frequent levels (clearance ↑), dose ↓ at delivery

Breastfeeding
Passes into milk (infant toxicity, hypothyroidism) → generally **avoid**, or specialist-monitored infant levels

✎ **CORRECTED** Original: "Teratogenic → use in first trimester and breastfeeding; drug of choice while pregnancy". Lithium is **avoided in the first trimester where possible** (Ebstein's anomaly) and is **not** the drug of choice in pregnancy; breastfeeding is generally not recommended. Seizures on lithium are a **sign of toxicity**, not a separate side effect.

2 Lithium Toxicity

Level (mEq/L)	Signs
1.5–2.5 (mild–moderate)	Vomiting, diarrhoea, **coarse tremor**, ataxia, slurred speech, drowsiness, confusion
> 2.5 (severe)	**Seizures**, coma, arrhythmias, renal failure, death

Management
- Stop lithium; **IV isotonic saline** (restore Na^+ & volume)
- **Haemodialysis** for severe toxicity / renal failure
- Seizures → benzodiazepines
- **Activated charcoal does NOT bind lithium**; whole-bowel irrigation for SR overdose

3 Drug & Food Interactions

↑ Lithium levels (toxicity)
- **Thiazide** diuretics (loop diuretics less)
- **NSAIDs** (except aspirin, sulindac)
- **ACEIs / ARBs**
- **Low-salt diet, dehydration**, vomiting, heat
- Elderly, unstable/reduced renal function
- Metronidazole

↓ Lithium levels (relapse)
- **Theophylline, caffeine**
- High sodium intake, sodium bicarbonate
- Osmotic diuretics, acetazolamide; pregnancy

4 Bipolar Disorder — Treatment Lines

ACUTE MANIA
- **Lithium** or **valproate**, or an **SGA** (quetiapine, aripiprazole, risperidone, olanzapine, asenapine, cariprazine)
- Severe: **Li / VPA + SGA**
- Stop antidepressants; short-term BZD for agitation

ACUTE BIPOLAR DEPRESSION
- **Quetiapine**, **lurasidone** (± Li/VPA), **lithium**, **lamotrigine**, cariprazine
- Olanzapine + fluoxetine combination
- **Avoid antidepressant monotherapy**

MAINTENANCE
- **Lithium** (first-line, prevents both poles) ± SGA
- **Quetiapine**, aripiprazole (mania prevention), valproate
- **Lamotrigine** — prevents **depressive** relapse (weak for mania)

Valproate must be avoided in women of child-bearing potential unless no alternative and a pregnancy-prevention programme is in place. **Lamotrigine** needs slow titration (rash/SJS). Based on CANMAT/ISBD 2018 and NICE CG185.

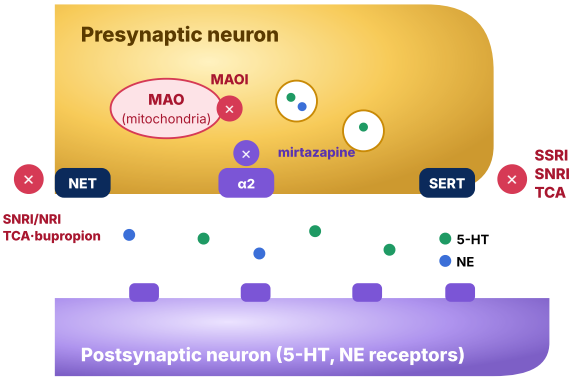
High-Yield Exam Pearls

- Lithium + **HCTZ / NSAID / ACEI** → toxicity.
- Lithium polyuria → **amiloride**; tremor → **propranolol**.
- Lithium in 1st trimester → **Ebstein's anomaly**.
- Bipolar depression maintenance → **lamotrigine**.

Major Depression & SSRIs

Pearson Review Notes

1 Monoamine Synapse — Where Antidepressants Act



Wk 1–2 Sleep, appetite, energy → **Wk 2–4** Mood begins to lift → **Wk 4–8** Judge response

- Assess response after an adequate trial of **4–8 weeks** at a therapeutic dose; if no improvement by ~4 weeks, adjust.
- Continue **≥ 6–12 months after remission** (first episode); longer if recurrent.
- Symptoms: low mood, **anhedonia**, sleep & appetite change, fatigue, guilt, poor concentration, **suicidal thoughts**.
- **Boxed warning**: ↑ suicidal thinking in patients **< 25 years** — monitor closely early on.
- Taper when stopping (discontinuation syndrome), except fluoxetine.

2 SSRIs — First-Line Antidepressants

selective serotonin reuptake inhibitors (block SERT)

Drug	Distinguishing points
Fluoxetine	**Longest half-life** (active metabolite norfluoxetine) → no taper needed; **wait 5 weeks** before starting an MAOI. Licensed for **children/adolescents**, **bulimia nervosa**, OCD, PMDD. Strong CYP2D6 inhibitor; activating (take in morning)
Sertraline	Preferred in **cardiac disease / post-MI**; commonly preferred in **pregnancy & breastfeeding**; GI upset/diarrhoea
Paroxetine	**Most anticholinergic & sedating**, weight gain; **worst discontinuation syndrome** (short t½); strong CYP2D6 inhibitor; **avoid in pregnancy** (cardiac malformations)
Fluvoxamine	**OCD**; **strong CYP1A2 & 2C19 inhibitor** (↑ clozapine, theophylline, tizanidine — CI with tizanidine, ramelteon)
Citalopram	**Dose-dependent QT prolongation**: max **40 mg/day**; max **20 mg/day** if > 60 years, hepatic impairment or CYP2C19 poor metaboliser
Escitalopram	S-enantiomer of citalopram; well tolerated, few interactions; QT risk (max 20 mg/day)
Dapoxetine	Short-acting SSRI for **premature ejaculation** (on-demand, 1–3 h before); not an antidepressant; syncope/orthostasis — CI with MAOIs, other SSRIs, cardiac disease

✎ **CORRECTED** Original: "Escitalopram → seizure" and "Paroxetine → use in epilepsy + depression". Neither is a recognised point: SSRIs (e.g. sertraline, citalopram, escitalopram) are generally safe in epilepsy at usual doses (avoid **bupropion**); paroxetine's key points are anticholinergic effects, discontinuation syndrome and **pregnancy risk**.

3 Managing SSRI Sexual Dysfunction

Wait / ↓ dose Some tolerance develops; lowest effective dose → **Switch** to **bupropion**, **mirtazapine**, vortioxetine or agomelatine → **Add-on** **Bupropion** augmentation; **PDE-5 inhibitor** (sildenafil) for erectile dysfunction → **Other** Cyproheptadine (sedating, may ↓ effect); ginseng — limited evidence

★ High-Yield Exam Pearls

- Adequate trial = **4–8 weeks**; treat ≥ 6–12 months after remission.
- Fluoxetine → MAOI: wait **5 weeks**; other SSRIs → **2 weeks**.
- Citalopram max **40 mg** (20 mg > 60 y) — QT.
- Post-MI depression → **sertraline**.

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SSRI Uses, Safety & Serotonin Syndrome

Pearson Review Notes

1 Indications

- Major depression
- Anxiety disorders: **GAD, panic**, social anxiety (first-line)
- **OCD** (higher doses), PTSD
- **Bulimia nervosa → fluoxetine**
- PMDD (fluoxetine, sertraline, paroxetine)
- Premature ejaculation (dapoxetine)

2 Adverse Effects

- **GI**: nausea, diarrhoea (early) → take **with/after food**
- **CNS**: agitation, insomnia (or somnolence), headache, tremor
- **Sexual dysfunction** (↓ libido, anorgasmia, delayed ejaculation)
- Weight change (paroxetine ↑)
- **Hyponatraemia / SIADH** — elderly, diuretics
- **↑ Bleeding** (↓ platelet 5-HT) with NSAIDs, aspirin, warfarin, DOACs
- **QT**: citalopram, escitalopram
- Suicidality < 25 y; activation/mania in bipolar

3 Serotonin Syndrome

Precipitants (with SSRIs/SNRIs)

- **MAOIs** (incl. linezolid, methylene blue, selegiline)
- **TCAs**, other SSRIs/SNRIs
- **Tramadol**, meperidine, fentanyl, methadone
- **Dextromethorphan**, triptans, lithium
- St John's wort, MDMA, buspirone

Features (Hunter criteria)

- Rapid onset (< 24 h)
- **Clonus** (inducible, ocular), **hyperreflexia**, tremor
- Agitation, confusion
- **Hyperthermia**, sweating, tachycardia, mydriasis
- **Diarrhoea**

Treatment

- **Stop all serotonergic drugs**
- Supportive care, IV fluids, cooling
- **Benzodiazepines** for agitation/rigidity
- **Cyproheptadine** (5-HT2A antagonist) for moderate cases
- Severe hyperthermia: sedation, paralysis, ICU

Discontinuation syndrome (≠ serotonin syndrome): dizziness, "electric-shock" sensations, irritability, flu-like symptoms after abrupt stop — **taper** over weeks (worst with paroxetine, venlafaxine).

4 Antidepressant Choice in Special Populations

Situation	Preferred	Avoid / note
Pregnancy	**Sertraline** (also citalopram/escitalopram, fluoxetine)	**Paroxetine** (cardiac defects); late-pregnancy SSRIs → neonatal adaptation, rare PPHN
Breastfeeding	**Sertraline**, paroxetine (low milk levels)	Fluoxetine (long t½, accumulates in infant)
Children / adolescents	**Fluoxetine** (≥ 8 y), escitalopram (≥ 12 y)	Monitor suicidality closely
Cardiac disease / post-MI	**Sertraline**	**TCAs**; high-dose citalopram (QT)
OCD	SSRIs e.g. **fluvoxamine**, fluoxetine, sertraline; TCA **clomipramine**	Higher doses & longer trials needed
Elderly	Sertraline, escitalopram, mirtazapine	TCAs, paroxetine (anticholinergic); watch Na^+, falls
Insomnia + weight loss	**Mirtazapine**	—
Obesity, fatigue, smoking, SSRI sexual SE	**Bupropion**	CI in seizures & eating disorders
Depression + neuropathic pain	**Duloxetine** (or TCA)	Duloxetine: avoid in hepatic disease

✎ **CORRECTED** Original listed "paroxetine" for pregnancy — it is the SSRI to **avoid in pregnancy**, though acceptable in breastfeeding.

High-Yield Exam Pearls

- Serotonin syndrome antidote → **cyproheptadine**.
- SSRI + linezolid / tramadol / MAOI → **serotonin syndrome**.
- Elderly on SSRI + diuretic → check **sodium**.
- SSRI + NSAID → add PPI (GI bleeding).

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SNRIs, NRIs & Atypical Antidepressants

Pearson Review Notes

1 SNRIs — Serotonin + Noradrenaline Reuptake Inhibitors

Venlafaxine
Low dose ≈ SSRI; NE effect at higher doses → **dose-related ↑ blood pressure** (monitor BP). Depression, GAD, panic, social anxiety. Severe discontinuation syndrome

Duloxetine
Diabetic peripheral neuropathy, fibromyalgia, chronic musculoskeletal pain, GAD. **Avoid in hepatic impairment / heavy alcohol use** and severe renal impairment (CrCl < 30)

Desvenlafaxine · Milnacipran
Desvenlafaxine = active metabolite of venlafaxine (renal adjust). Milnacipran: fibromyalgia

Side effects ≈ SSRIs (nausea, sexual dysfunction, hyponatraemia, bleeding, serotonin syndrome) **+ noradrenergic**: ↑ BP, ↑ HR, sweating, dry mouth, constipation, urinary hesitancy.

2 Selective Noradrenaline Reuptake Inhibitors (NRIs)

Reboxetine
Antidepressant (not available in US); act only on NE. Insomnia, sweating, urinary hesitancy

Atomoxetine
Non-stimulant for **ADHD**; CYP2D6 substrate; boxed warning for suicidal ideation; rare hepatotoxicity

3 Atypical Antidepressants

Bupropion — NDRI

MOA NE & DA reuptake inhibitor; **no serotonergic activity**

Uses Depression with **fatigue / obesity**; to avoid or treat **SSRI sexual dysfunction**; **smoking cessation**; seasonal affective disorder

SE Insomnia, agitation, dry mouth, tremor, weight loss, **dose-related seizures**

CI **Seizure disorder**; **bulimia/anorexia** (electrolyte imbalance ↑ seizure risk); abrupt alcohol/BZD/antiepileptic withdrawal; MAOI within 14 days

Smoking cessation options: **varenicline** (most effective single agent), bupropion, NRT; combination NRT is also very effective.

Mirtazapine — NaSSA

MOA **α2-antagonist** (↑ release of NE & 5-HT) + 5-HT2 & 5-HT3 antagonist + strong **H1 antagonist**

Uses Depression with **insomnia, poor appetite / weight loss** ("depressed + thin"), elderly; few sexual SEs, little nausea

SE **Sedation** (more at low dose), ↑ appetite & **weight gain**, dry mouth; rare agranulocytosis

✎ **CORRECTED** Original: mirtazapine = "serotonin & NE reuptake inhibitor". It is **not** a reuptake inhibitor — it is an **α2-antagonist**.

Trazodone
5-HT2A antagonist + weak SERT inhibition; low dose used off-label for **insomnia**. SE: sedation, orthostasis, **priapism** (urologic emergency)

Vortioxetine · Agomelatine
Vortioxetine: multimodal 5-HT agent, nausea, fewer sexual SEs. Agomelatine: MT1/MT2 agonist + 5-HT2C antagonist — **LFT monitoring** (hepatotoxicity)

High-Yield Exam Pearls

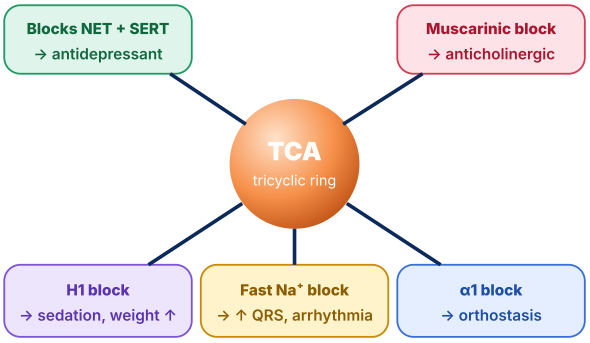
- Bupropion: **no sexual SE, weight loss, but seizures**.
- Mirtazapine: **sleep + appetite**.
- Venlafaxine → monitor **blood pressure**.
- Trazodone → **priapism**.

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Tricyclic Antidepressants (TCAs)

Pearson Review Notes

1 Mechanism — One Drug, Many Receptors



- Block reuptake of **NE and 5-HT**; also antagonise **muscarinic, H1, α1** receptors and cardiac **Na⁺ channels** (quinidine-like, class Ia effect).
- Long half-life (once-daily bedtime dosing) and **narrow therapeutic index**.
- Second-line for depression because of side effects and overdose toxicity.

Tertiary amines	Secondary amines
Amitriptyline, imipramine, clomipramine, doxepin — **more anticholinergic & sedating**	Nortriptyline, desipramine — metabolites; **better tolerated** (preferred in elderly if a TCA is needed)

✎ **CORRECTED** "Imipramine (more potent)" removed — potency differences are not a recognised exam point; the key distinction is tertiary vs secondary amine tolerability.

2 Uses

- Moderate–severe / refractory depression
- **Neuropathic pain** (amitriptyline, nortriptyline)
- **Migraine & tension headache prophylaxis** → amitriptyline
- **Insomnia** (sleep maintenance) → **low-dose doxepin** (3–6 mg)
- **OCD** → **clomipramine** (most serotonergic TCA)
- Panic disorder (imipramine); **nocturnal enuresis** (imipramine)

3 Adverse Effects

- **Anticholinergic**: dry mouth, constipation, urinary retention, blurred vision, tachycardia, confusion (elderly)
- **Orthostatic hypotension** (falls), sedation, weight gain
- **Cardiotoxicity**: ↑ PR/QRS/QT, life-threatening arrhythmias (quinidine-like)
- ↓ Seizure threshold; sexual dysfunction
- Photosensitivity; rare **blue-green urine** (amitriptyline)

4 TCA Overdose — "3 Cs"

Coma
CNS depression

Convulsions
Seizures → BZDs

Cardiotoxicity
QRS > 100 ms, arrhythmia, hypotension

Antidote: IV sodium bicarbonate (alkalinisation + Na⁺ load overcomes Na⁺-channel block) when QRS is wide, arrhythmia or hypotension. Activated charcoal if early & airway protected. **Avoid physostigmine & flumazenil** (seizure/asystole risk).

Contraindications / Cautions

- Recent **myocardial infarction**, heart block, arrhythmia
- **Narrow-angle glaucoma**
- **BPH / urinary retention**, severe constipation
- Epilepsy (caution), elderly (Beers list)
- **MAOIs** (within 14 days) → serotonin syndrome / hypertensive crisis
- Other QT drugs; CYP2D6 inhibitors (fluoxetine, paroxetine) ↑ TCA levels

5 Individual TCAs

Drug	Type	Characteristic use / point
Amitriptyline	Tertiary	Neuropathic pain, migraine prophylaxis; very sedating; → nortriptyline
Nortriptyline	Secondary	Less orthostasis & anticholinergic effect; level monitoring possible
Imipramine	Tertiary	Nocturnal enuresis, panic disorder; metabolised to desipramine
Clomipramine	Tertiary	OCD (most serotonergic); seizure risk
Doxepin	Tertiary	Low dose (3–6 mg) for sleep maintenance; topical for pruritus

High-Yield Exam Pearls

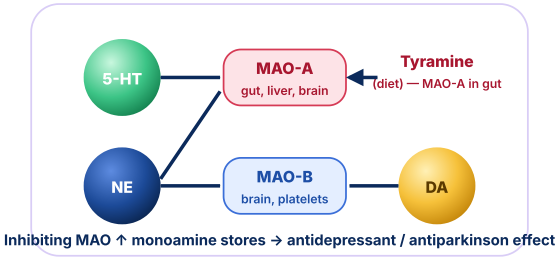
- TCA overdose → wide QRS → **sodium bicarbonate**.
- Migraine prophylaxis + insomnia → **amitriptyline**.
- OCD TCA → **clomipramine**; enuresis → **imipramine**.
- Elderly → prefer **nortriptyline** if a TCA is essential.

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Monoamine Oxidase Inhibitors (MAOIs)

Pearson Review Notes

1 Mechanism & Classification



Type	Drugs	Use
Non-selective, irreversible (A + B)	Phenelzine, tranylcypromine, isocarboxazid	Refractory / atypical depression
Selective MAO-A, reversible (RIMA)	Moclobemide	Depression; lower tyramine risk
Selective MAO-B, irreversible	Selegiline, rasagiline (safinamide: reversible)	**Parkinson's disease**; selegiline patch for depression

✎ **CORRECTED** Original: MAO breaks down "catecholamines like 5-HT" and MAO-B is "only found in brain". 5-HT is an **indolamine**, not a catecholamine (MAO metabolises **monoamines**: 5-HT, NE, DA); MAO-B is also present in **platelets** and other tissues.

2 Hypertensive Crisis — the "Cheese Reaction"

Tyramine-rich food
Aged cheese, cured/smoked meat, soy sauce, fermented foods, tap beer, some wines. **Avoid during & 2 weeks after** an irreversible MAOI

→

Gut MAO-A blocked
Tyramine absorbed intact → displaces NE from nerve terminals

→

Massive NE release
Severe hypertension, occipital headache, palpitations, stroke risk

→

Treat
Phentolamine IV (α-blocker); alternatives nicardipine, labetalol

3 Adverse Effects

- **Orthostatic hypotension** (most common), dizziness
- Insomnia, agitation, headache, nausea, weight gain, oedema, sexual dysfunction
- Hypertensive crisis & serotonin syndrome (interactions)
- Hallucinations/confusion — mainly selegiline + levodopa in Parkinson's

4 Washout Rules & Dangerous Combinations



- SSRIs/SNRIs/TCAs, **linezolid**, methylene blue, **meperidine**, tramadol, **dextromethorphan**, St John's wort → serotonin syndrome
- **Sympathomimetics** (pseudoephedrine, phenylephrine, amphetamines), levodopa, bupropion → hypertensive crisis

5 Antidepressant Classes — Quick Comparison

Class	Sexual SE	Weight	Sedation	Special risk
SSRIs	High	± (paroxetine ↑)	Low	Bleeding, ↓Na^+, QT (citalopram)
SNRIs	High	Neutral	Low	↑ BP (venlafaxine), liver (duloxetine)
Bupropion	Very low	↓	None (activating)	Seizures
Mirtazapine	Low	↑↑	High	Agranulocytosis (rare)
TCAs	Moderate	↑	High	Cardiotoxic, lethal overdose
MAOIs	Moderate	↑	Variable	Tyramine hypertensive crisis

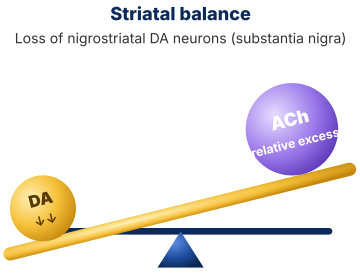
High-Yield Exam Pearls

- MAOI + aged cheese → hypertensive crisis → **phentolamine**.
- Washout **2 weeks** (5 weeks after fluoxetine).
- MAO-B inhibitors (**selegiline, rasagiline**) → Parkinson's.
- MAOI + pseudoephedrine cold remedy = **contraindicated**.

Parkinson's Disease & Levodopa

Pearson Review Notes

1 Pathophysiology — Dopamine / Acetylcholine Imbalance



T — Tremor at rest
"pill-rolling", unilateral onset

R — Rigidity
cogwheel / lead-pipe

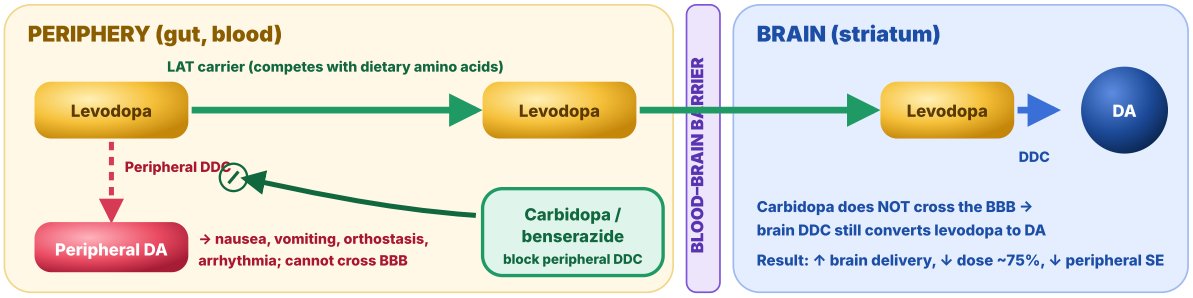
A — Akinesia / bradykinesia
slow, reduced movement

P — Postural instability
shuffling gait, falls (later)

Non-motor: depression, constipation, anosmia, sleep disorders, dementia, orthostatic hypotension.

Strategy: ↑ dopamine signalling (levodopa, agonists, MAO-B & COMT inhibitors) or ↓ acetylcholine (anticholinergics).

2 Why Levodopa Is Always Given With Carbidopa / Benserazide



Dopamine itself cannot cross the BBB, so its precursor levodopa is used. Usual levodopa : carbidopa ratios are 4:1 or 10:1; ≈ 70–100 mg/day of carbidopa is needed to saturate peripheral DDC.

✎ **CORRECTED** Original: "the dose of levodopa depends on the carbidopa dose". The levodopa dose is titrated to symptoms; carbidopa simply needs to reach an adequate daily amount (≈ 70–100 mg) to block peripheral conversion.

3 Antiparkinsonian Drug Classes at a Glance

Levodopa / carbidopa	DA agonists	MAO-B inhibitors	COMT inhibitors	Anticholinergics	Amantadine
Most effective; DA precursor	Pramipexole, ropinirole, rotigotine, apomorphine	Selegiline, rasagiline, safinamide	Entacapone, opicapone, tolcapone	Benztropine, trihexyphenidyl, biperiden	NMDA antagonist; ↓ dyskinesia

4 Levodopa Adverse Effects

Peripheral (↓ by carbidopa)
- Nausea, vomiting, anorexia
- **Orthostatic hypotension**
- Cardiac arrhythmias
- Dark urine / sweat (harmless)

Central
- **Dyskinesias** (peak-dose involuntary movements)
- **Hallucinations**, psychosis, confusion
- Insomnia, vivid dreams, depression
- Impulse-control disorders (less than agonists)

★ High-Yield Exam Pearls

- TRAP: **Tremor, Rigidity, Akinesia, Postural instability**.
- Carbidopa blocks **peripheral** DOPA decarboxylase only.
- Levodopa = **most effective** symptomatic drug.
- Nausea on levodopa → give **domperidone**, never metoclopramide.

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Levodopa: Use, Interactions & Fluctuations

Pearson Review Notes

1 How to Take Levodopa

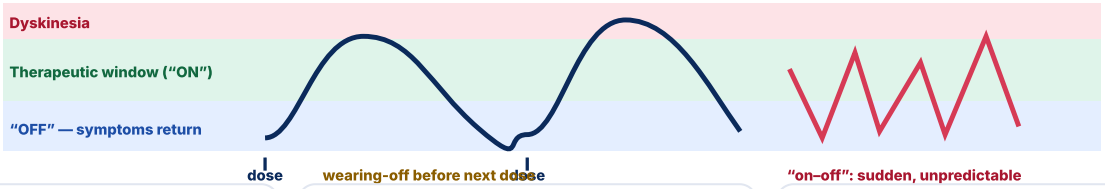
- Best on an **empty stomach**: ~30 min before or 1 h after meals
- **High-protein meals** compete for intestinal & BBB amino-acid (LAT) transport → ↓ effect: spread protein through the day or shift it to the evening
- If nausea: take with a **small low-protein snack** (or extra carbidopa)
- Space doses regularly; never stop abruptly (risk of **NMS-like hyperpyrexia**)

2 Drug & Food Interactions

Interacting agent	Effect	Advice
Non-selective MAOIs (phenelzine, tranylcypromine)	**Hypertensive crisis**	**Contraindicated** — stop MAOI ≥ 2 weeks before
MAO-B & COMT inhibitors	**Prolong & enhance** levodopa effect ("on" time ↑)	Intended combination; may need to **↓ levodopa dose** (dyskinesia)
Dopamine antagonists: metoclopramide, prochlorperazine, haloperidol, risperidone	Block DA → worsen parkinsonism	**Avoid**. For nausea use **domperidone** (does not cross BBB; QT caution); for psychosis use quetiapine/clozapine
Pyridoxine (vitamin B6)	Cofactor that **↑ peripheral decarboxylation** → ↓ brain levodopa	Relevant only with levodopa **alone**; carbidopa abolishes this
Iron (ferrous salts)	Chelation → ↓ absorption	Separate by ≥ 2 hours
Antihypertensives	Additive orthostatic hypotension	Monitor BP, may ↓ antihypertensive dose

✎ **CORRECTED** Original: "Vitamin B6 ↓ DDC activity" — B6 is the **cofactor that increases** it; "MAO-B/COMT increase the frequency of levodopa" — they **extend** each dose's effect; the PPI "2 h before / 4 h after" rule is not established (antacid/PPI effects are minor and inconsistent).

3 Motor Complications With Long-Term Levodopa



Wearing-off (end-of-dose)

Effect fades before next dose → **↑ dose frequency / shorter interval**, CR formulation, or add **COMT inhibitor**, **MAO-B inhibitor** or **DA agonist**

"On–off" phenomenon

Sudden unpredictable switches after long good control → **adjust regimen**: add COMT/MAO-B inhibitor or agonist; **apomorphine SC rescue**; advanced: infusion therapy or DBS. **Do not stop levodopa**

Peak-dose dyskinesia

Too much DA at peak → ↓ individual doses (smaller, more frequent), add **amantadine**

✎ **CORRECTED** Original: on–off → "discontinue levodopa and consider other treatments". Levodopa is **never withdrawn** for fluctuations (abrupt withdrawal risks a parkinsonism-hyperpyrexia/NMS-like syndrome); the regimen is optimised instead.

High-Yield Exam Pearls

- Levodopa + **phenelzine** = hypertensive crisis.
- Levodopa-induced dyskinesia → **amantadine**.
- Wearing-off → add **entacapone** / shorten interval.
- Protein meals ↓ levodopa effect.

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Dopamine Agonists, MAO-B & COMT Inhibitors

Pearson Review Notes

1 Dopamine Agonists — act directly on D2/D3 receptors

Ergot agonists — bromocriptine, cabergoline
Rarely used for PD now because of **fibrotic reactions** (pulmonary, retroperitoneal, **cardiac valvulopathy** — echo monitoring). **Cabergoline** is better tolerated and long-acting (twice weekly) → **first choice for hyperprolactinaemia**.

Non-ergot agonists — pramipexole, ropinirole, rotigotine, apomorphine
Preferred in PD. Pramipexole: high oral bioavailability, **renal** elimination (dose adjust). Ropinirole: CYP1A2. Rotigotine: transdermal patch. **Apomorphine SC**: rescue for "off" episodes (give antiemetic).

Uses

- Parkinson's disease (early monotherapy in younger patients; adjunct)
- **Hyperprolactinaemia** / prolactinoma, lactation suppression (cabergoline, bromocriptine)
- **Restless legs syndrome** (pramipexole, ropinirole, rotigotine)
- **NMS** (bromocriptine); **type 2 diabetes** (bromocriptine QR, US); acromegaly

Adverse effects

- Nausea, **orthostatic hypotension**, peripheral oedema
- **Hallucinations**, confusion (more than levodopa, esp. elderly)
- **Sudden sleep attacks** / somnolence — warn about driving
- **Impulse-control disorders**: gambling, hypersexuality, shopping
- Ergots: pulmonary fibrosis, valvulopathy

Apomorphine is contraindicated with 5-HT3 antagonists (ondansetron, granisetron) → profound hypotension & loss of consciousness; also contains a **sulfite** (sulfite allergy). Use domperidone for nausea. *(Original note attached these CIs to all agonists — they apply to apomorphine.)*

2 Drugs That Change Prolactin

↑ Prolactin	↓ Prolactin
D2 antagonists: **antipsychotics** (risperidone), **metoclopramide**, domperidone; **methyldopa**, **reserpine**; **cimetidine**; estrogens; opioids; TCAs/SSRIs (mild)	**Dopamine agonists**: bromocriptine, cabergoline; **levodopa**

✎ **CORRECTED** "Androgens / spironolactone" removed from the ↓ prolactin list — spironolactone causes gynaecomastia through **anti-androgen** activity, not prolactin. Ergotamine is not used to lower prolactin.

Memory aid — drugs that ↑ liver enzymes: tolcapone (CNS), methotrexate (immune), glitazones (troglitazone withdrawn; check LFTs with pioglitazone).

3 MAO-B & COMT Inhibitors

Drug	Key points
Selegiline	Irreversible MAO-B; metabolised to **L-amphetamine** → **insomnia**, give in the **morning**
Rasagiline	Irreversible MAO-B, once daily, **no amphetamine metabolites**; CYP1A2 (↓ dose with ciprofloxacin)
Entacapone	Peripheral COMT inhibitor taken **with each levodopa dose**; diarrhoea, dyskinesia, **orange-brown urine**; no LFT monitoring
Opicapone	Once-daily COMT inhibitor at bedtime
Tolcapone	Central + peripheral; **boxed warning: fulminant hepatotoxicity** → LFT monitoring, last-line

MAO-B inhibitors interact with meperidine, tramadol, methadone, dextromethorphan, SSRIs/SNRIs (serotonin syndrome).

4 Levodopa vs Dopamine Agonists

Feature	Levodopa / carbidopa	Non-ergot DA agonists
Motor efficacy	**Highest**	Lower
Long-term motor complications (dyskinesia, fluctuations)	More frequent	**Fewer** (delay levodopa)
Hallucinations, confusion	Less	**More** (avoid in elderly / cognitive impairment)
Sleep attacks, impulse-control disorders, oedema	Less	**More**
Typical patient	Older, disabling symptoms	Younger, early disease

★ High-Yield Exam Pearls

- Hyperprolactinaemia → **cabergoline**.
- Restless legs → **pramipexole / ropinirole**.
- Tolcapone → **hepatotoxicity**; entacapone → orange urine.
- Apomorphine + **ondansetron** = contraindicated.

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Parkinson's: Adjuncts & Drug Summary

Pearson Review Notes

1 Anticholinergics & Amantadine

- **Benztropine, trihexyphenidyl, biperiden**: mainly for **tremor** in younger patients; also drug-induced EPS
- **Avoid in elderly** — confusion, memory loss, urinary retention, constipation, glaucoma (Beers)
- **Amantadine**: antiviral (influenza A — no longer recommended) + **NMDA antagonist**, ↑ DA release → mild PD & **levodopa-induced dyskinesia**
- SE: **livedo reticularis**, ankle oedema, hallucinations, insomnia; renal dose adjustment

✎ **CORRECTED** Amantadine is not a central anticholinergic; "pigmentation" belongs to COMT inhibitors (urine discolouration).

2 Choosing Initial Therapy

NICE NG71 / AAN

Symptoms impair life
start treatment

→

Younger / mild
MAO-B inhibitor or DA agonist (or levodopa)

Older / disabling / cognitive issues
Levodopa/carbidopa (best efficacy, fewer psychiatric SE)

→

Later
add COMT / MAO-B inhibitor, agonist or amantadine

3 Antiparkinsonian Drugs — Summary Table

Class	Examples	Mechanism	Key adverse effects	Exam pearl
DA precursor	Levodopa + carbidopa / benserazide	Converted to DA in brain; carbidopa blocks peripheral DDC	Nausea, orthostasis, dyskinesia, hallucinations	Most effective; protein meals ↓ effect
DA agonists (non-ergot)	Pramipexole, ropinirole, rotigotine, apomorphine	Direct D2/D3 stimulation	Sleep attacks, impulse control, hallucinations, oedema	RLS; apomorphine × ondansetron
DA agonists (ergot)	Bromocriptine, cabergoline	Direct D2 stimulation	Fibrosis, valvulopathy	Cabergoline → hyperprolactinaemia
MAO-B inhibitors	Selegiline, rasagiline, safinamide	↓ DA breakdown in brain	Insomnia (selegiline), serotonin syndrome DDIs	Avoid meperidine, tramadol
COMT inhibitors	Entacapone, opicapone, tolcapone	↓ peripheral levodopa breakdown → longer effect	Diarrhoea, dyskinesia, orange urine; tolcapone liver	Only useful **with levodopa**; wearing-off
Anticholinergics	Benztropine, trihexyphenidyl, biperiden	Block central muscarinic receptors	Confusion, retention, dry mouth	Tremor in young; avoid elderly
NMDA antagonist	Amantadine	↑ DA release, ↓ glutamate	Livedo reticularis, oedema	Levodopa-induced dyskinesia

Drugs That Worsen Parkinsonism

- Antipsychotics (haloperidol, risperidone)
- Metoclopramide, prochlorperazine, promethazine
- Reserpine, tetrabenazine, methyldopa
- Valproate, lithium (tremor)

Safer Choices in Parkinson's

- Nausea → **domperidone**, ondansetron (not with apomorphine)
- Psychosis → **quetiapine, clozapine, pimavanserin**
- Dementia → **rivastigmine**
- Orthostasis → review drugs; fludrocortisone, midodrine

High-Yield Exam Pearls

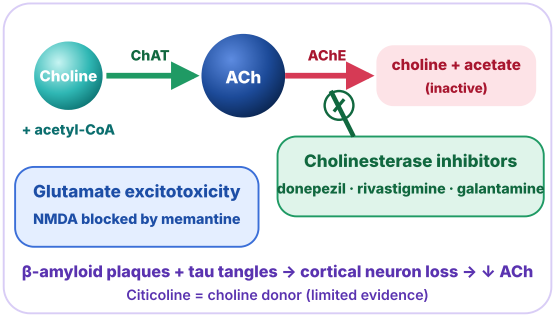
- COMT inhibitors work only **with levodopa**.
- Amantadine → **livedo reticularis**.
- Elderly PD patient → avoid **anticholinergics**.
- PD dementia → **rivastigmine** (licensed).

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Alzheimer's Disease (AD)

Pearson Review Notes

1 Pathophysiology & Cholinergic Targets



- AD is the **most common cause of dementia**; neurodegeneration of the cerebral cortex & hippocampus.
- Late-onset ("senile") AD: age ≥ 65 years.
- Drugs are **symptomatic**: modest benefit in cognition and function; reassess regularly.
- **Anti-amyloid mAbs** (lecanemab, donanemab) for early AD/MCI with confirmed amyloid — risk of **ARIA** (oedema/haemorrhage), MRI monitoring.

✎ **CORRECTED** Original: "Dementia + Alzheimer → only rivastigmine". AD is itself a dementia; **rivastigmine** is the agent licensed for **Parkinson's disease dementia**.

2 Drug Therapy

Drug	Stage	Key points
Donepezil	Mild, moderate & severe	AChE inhibitor; once daily 5 → 10 mg; **23 mg/day** for moderate–severe after ≥ 3 months on 10 mg. Usually at **bedtime** (morning if vivid dreams/insomnia)
Rivastigmine	Mild–moderate (patch also severe)	Inhibits AChE + butyrylcholinesterase; capsules (take with food) or **transdermal patch** (4.6 → 9.5 → 13.3 mg/24 h) — **fewer GI effects**; PD dementia
Galantamine	Mild–moderate	AChE inhibitor + nicotinic modulator; with food; renal/hepatic dose limits
Memantine	Moderate–severe	**NMDA receptor antagonist**; alone or **with donepezil**; dizziness, confusion, headache, constipation; ↓ dose if CrCl < 30

ChEI adverse effects: nausea, vomiting, diarrhoea, anorexia/weight loss (titrate slowly); **bradycardia, syncope** (caution with β-blockers, sick sinus); insomnia/vivid dreams; muscle cramps; urinary incontinence.

Herbal & supplements: Ginkgo biloba — no proven benefit, ↑ **bleeding** with anticoagulants/antiplatelets. Vitamin E — limited data, bleeding at high doses. Caprylidene (medical food) — weak evidence. Vitamin D — no proven benefit.

✎ **CORRECTED** Original: "advanced → only donepezil 23 mg". Moderate–severe AD: **memantine**, donepezil (10 or 23 mg), or **donepezil + memantine**; rivastigmine 13.3 mg patch is also licensed for severe AD. Memantine's main action is **NMDA** antagonism (5-HT3 / nicotinic effects are minor).

Q. A 79-year-old man with mild-to-moderate AD is on oral rivastigmine 6 mg twice daily but cannot tolerate it because of nausea and vomiting. Best recommendation?

A) Prochlorperazine suppositories 25 mg 60 min before each dose
B) Stop rivastigmine and substitute galantamine 4 mg twice daily
C) Metoclopramide 10 mg 30 min before each dose
D) Change to transdermal rivastigmine 4.6 mg/24 h

Answer: D. The patch gives steadier levels and far fewer GI effects. A and C add **D2 antagonists** (EPS, confusion; antipsychotic-type risk in dementia). Note: per label, 6–12 mg/day oral converts to 9.5 mg/24 h; restarting at 4.6 mg is reasonable after intolerance. Apply the first patch the day after the last oral dose.

Comorbidity Traps

- **Overactive bladder + AD**: avoid **oxybutynin** (crosses BBB, anticholinergic, opposes ChEI) → prefer **mirabegron**, vibegron or trospium
- Avoid anticholinergics: diphenhydramine, TCAs, paroxetine
- Antipsychotics only for severe agitation (boxed warning)

✎ **CORRECTED** Original recommended **oxybutynin** for urinary urgency + AD — this is a classic error.

High-Yield Exam Pearls

- ChEI GI intolerance → switch to **rivastigmine patch**.
- Moderate–severe → add **memantine** (NMDA).
- ChEIs → watch **bradycardia/syncope**.
- Never pair a ChEI with **oxybutynin**.

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Epilepsy: Basics & Seizure Types

Pearson Review Notes

1 Definitions

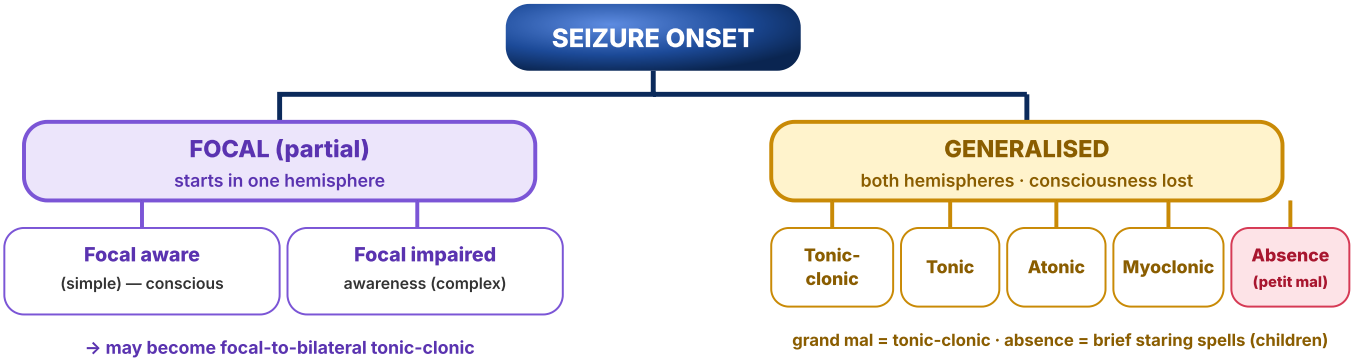
Seizure
A single episode of abnormal, excessive, synchronous neuronal **electrical** activity in the brain

Epilepsy
≥ 2 unprovoked seizures > 24 h apart, or 1 unprovoked seizure with high recurrence risk (ILAE)

Status epilepticus
Seizure lasting **≥ 5 minutes**, or repeated seizures **without recovery** in between → **emergency**

✎ **CORRECTED** Original: "abnormal *electrolyte* activity" → abnormal **electrical** activity.

2 ILAE Seizure Classification

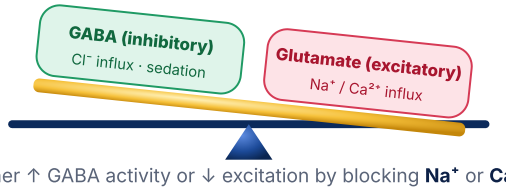


Absence seizures: first-line **ethosuximide** (T-type Ca^{2+} channel blocker in thalamus) or **valproate** (if GTC seizures coexist); lamotrigine is an alternative.

May worsen absence/myoclonic seizures: carbamazepine, oxcarbazepine, phenytoin, gabapentin, pregabalin, vigabatrin, tiagabine.

✎ **CORRECTED** Original: absence → "only ethosuximide". Ethosuximide **or valproate** are first-line (valproate preferred when tonic-clonic seizures also occur, but avoided in women of child-bearing potential).

3 Excitation vs Inhibition



ASMs either ↑ GABA activity or ↓ excitation by blocking **Na^+** or **Ca^{2+}** channels, glutamate receptors or modulating SV2A.

4 Seizure First Aid

DO
- **Time** the seizure; stay calm
- Protect the head; move hard/sharp objects away
- Loosen tight clothing
- After jerking stops → **recovery (side) position**

DON'T
- **Never put anything in the mouth**
- Don't restrain movements
- No food/drink until fully alert

Call emergency if **> 5 min**, repeated, first seizure, injury, pregnancy or breathing difficulty.

✎ **CORRECTED** Original: "open the mouth" — **unsafe** (dental/airway injury).

High-Yield Exam Pearls

- Status epilepticus = **≥ 5 min** or no recovery between seizures.
- Absence → **ethosuximide** (or valproate).
- **Ketogenic diet** (high-fat, very low-carbohydrate) → refractory paediatric epilepsy.
- Carbamazepine/phenytoin can **worsen absence** seizures.

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Status Epilepticus & Antiseizure Drug Map

Pearson Review Notes

1 Convulsive Status Epilepticus — Time-Based Algorithm

AES 2016 guideline (adults)

0–5 min · Stabilise
ABC, O_2, monitor, IV access; **check glucose** (if low: thiamine 100 mg IV then dextrose 50%); time the seizure

→

5–20 min · Benzodiazepine
IV lorazepam 0.1 mg/kg (max 4 mg/dose) *or* **IM midazolam** 10 mg (> 40 kg) *or* IV diazepam 0.15–0.2 mg/kg (max 10 mg); may repeat once. No IV: rectal diazepam, buccal/intranasal midazolam

→

20–40 min · Second-line
One of: **fosphenytoin** 20 mg PE/kg (max 1500 mg PE) · **levetiracetam** 60 mg/kg (max 4500 mg) · **valproate** 40 mg/kg (max 3000 mg); phenobarbital 15 mg/kg if none available

→

40+ min · Refractory
ICU, intubation, continuous EEG; anaesthetic infusion: **midazolam, propofol**, pentobarbital / thiopental

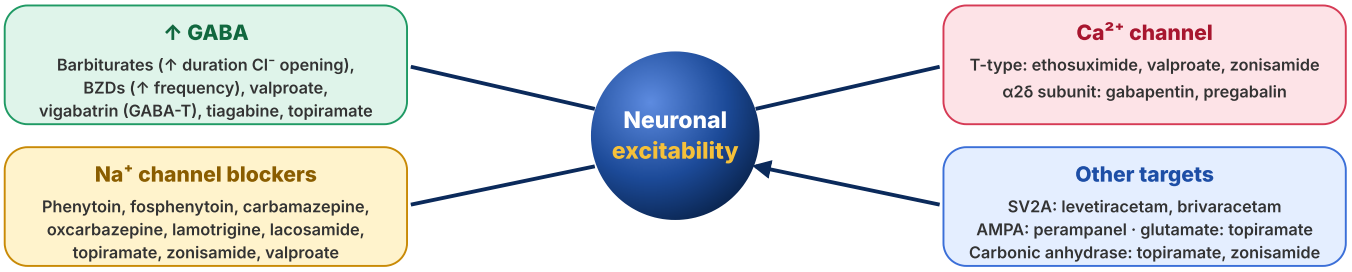
Pre-hospital: **buccal/intranasal midazolam** or **rectal diazepam**.

Fosphenytoin max rate 150 mg PE/min; phenytoin 50 mg/min (hypotension, arrhythmia). IV clonazepam is an alternative where available.

Levetiracetam/valproate equally effective as fosphenytoin (ESETT). **Avoid valproate** in liver disease, pregnancy.

✎ **CORRECTED** Original: emergency treatment "IV clonazepam" and "muscle spasm → diazepam". First-line is a **benzodiazepine — IV lorazepam / IM midazolam / IV diazepam**; diazepam's muscle-relaxant use is separate from status epilepticus.

2 Antiseizure Medicines by Mechanism



✎ **CORRECTED** Original grouped **gabapentin & pregabalin** with GABA-enhancing drugs. Despite being GABA analogues, they **do not act on GABA receptors**; they bind the **α2δ subunit** of voltage-gated Ca^{2+} channels.

3 Older vs Newer Generation ASMs

Generation	Drugs	Characteristics
Older	Phenobarbital, primidone, phenytoin, carbamazepine, valproate, ethosuximide	Effective; more adverse effects, **enzyme induction/inhibition** and drug interactions, teratogenicity; level monitoring often needed
Newer	Lamotrigine, levetiracetam, oxcarbazepine, topiramate, lacosamide, zonisamide, gabapentin, pregabalin, vigabatrin, brivaracetam, perampanel	Similar efficacy, generally **better tolerability** and fewer interactions. Several are used as **monotherapy** (lamotrigine, levetiracetam, oxcarbazepine, topiramate, lacosamide); others adjunctive

✎ **CORRECTED** Original: newer ASMs = "combination only, except lamotrigine". Outdated — many newer agents are licensed / guideline-recommended as monotherapy.

High-Yield Exam Pearls

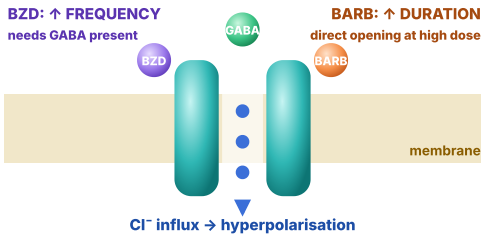
- First-line status: **IV lorazepam** or **IM midazolam**.
- Second-line: **fosphenytoin = levetiracetam = valproate**.
- Gabapentinoids → **α2δ Ca^{2+}**, not GABA receptors.
- Barbiturates ↑ **duration**, BZDs ↑ **frequency** of Cl^- channel opening.

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Barbiturates & Benzodiazepines

Pearson Review Notes

1 GABA-A Receptor — Two Ways to Open the Chloride Channel



Why BZDs replaced barbiturates
✔ **Wider therapeutic index** (ceiling effect — barbiturates open channels directly at high dose)
✔ More selective; less CNS, CV & **respiratory depression**; less tolerance & dependence
✔ **Antidote available** (flumazenil)
✔ **No enzyme induction** (barbiturates are strong CYP inducers)

2 Barbiturates

Long-acting
Phenobarbital (epilepsy incl. neonatal, status), **primidone** (prodrug → phenobarbital + PEMA; essential tremor)

Short / intermediate
Amobarbital, butalbital (headache combos), pentobarbital (refractory status), secobarbital — sedatives, now rarely used

Ultra-short
Thiopental, methohexital — IV anaesthesia induction (rapid redistribution)

Adverse effects: sedation, cognitive impairment, paradoxical hyperactivity (children), **CNS, CV & respiratory depression** (fatal with alcohol/opioids), tolerance, dependence, withdrawal seizures, skin rash (SJS rare), osteomalacia.

Interactions/CI: potent **CYP inducer** (↓ OCs, warfarin, many drugs). **CI in acute intermittent porphyria**. No antidote — supportive care; urinary alkalinisation ↑ phenobarbital excretion.

✎ **CORRECTED** Original classed secobarbital as intermediate-acting; it is **short-acting**.

3 Benzodiazepines

Duration	Drugs	Notes
Long	Diazepam, chlordiazepoxide, clonazepam, flurazepam	Active metabolites → **accumulation**/hangover (elderly, liver disease). Alcohol withdrawal (chlordiazepoxide, diazepam); clonazepam: panic, seizures
Intermediate	**Lorazepam, oxazepam, temazepam** ("LOT"); alprazolam	**Glucuronidation only, no active metabolites** → preferred in **elderly & hepatic impairment**; lorazepam for alcohol withdrawal with liver disease, status epilepticus
Short	Midazolam, triazolam	Procedural sedation, insomnia (triazolam); more **rebound insomnia/anxiety**, dependence & withdrawal

Adverse effects
- Drowsiness, confusion, **falls** (elderly — Beers: avoid)
- **Anterograde amnesia** (can't form new memories)
- **Respiratory depression** — boxed warning with **opioids**
- Tolerance, dependence; **withdrawal seizures** if stopped abruptly → taper
- Rebound insomnia; paradoxical agitation

Antidote & special groups
- **Flumazenil IV** — competitive BZD antagonist; short-acting (re-sedation); **may precipitate seizures** in dependence or TCA co-ingestion
- **Pregnancy**: avoid if possible (floppy infant, neonatal withdrawal); **breastfeeding**: avoid long-acting **diazepam** (lipophilic, accumulates)

✎ **CORRECTED** Chlordiazepoxide is **long-acting** (not intermediate); in the elderly BZDs are best avoided — if essential, use an intermediate "LOT" agent, not a short-acting one.

★ High-Yield Exam Pearls

- Elderly / liver disease BZD → **lorazepam, oxazepam, temazepam**.
- BZD overdose → **flumazenil** (caution seizures).
- Phenobarbital = **enzyme inducer**; CI porphyria.
- BZD + opioid → **fatal respiratory depression**.

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Valproate & Gabapentinoids

Pearson Review Notes

1 Valproic Acid / Sodium Valproate — Broad Spectrum

Mechanism & uses

- Na^+ channel block + T-type Ca^{2+} block + ↑ **GABA**
- Effective in **all seizure types**: GTC, absence, myoclonic, focal
- **Bipolar mania**; **migraine prophylaxis**
- Therapeutic level ≈ **50–100 mcg/mL** (mania up to ~125)

Boxed warnings

- **Hepatotoxicity** — highest risk < 2 years, polytherapy, mitochondrial disease (POLG) → LFTs
- **Pancreatitis** (abdominal pain, vomiting)
- **Teratogenicity**: neural-tube defects (**spina bifida**), other malformations, ↓ IQ, autism

Other adverse effects

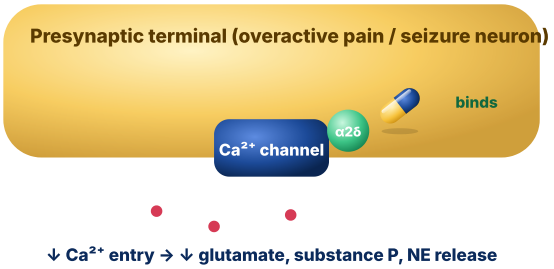
- Weight gain, tremor, **alopecia** (hair thinning)
- **Thrombocytopenia** (dose-related)
- **Hyperammonaemia** / encephalopathy → **L-carnitine**
- GI upset, PCOS-like effects, hyponatraemia

Pregnancy: avoid in girls & women of child-bearing potential unless no alternative, with a **pregnancy-prevention programme** (MHRA/EMA). High-dose folic acid does **not** remove the risk.

Interactions: enzyme **inhibitor** → ↑ lamotrigine (halve dose, ↑ SJS), ↑ phenobarbital; **carbapenems ↓ valproate levels dramatically** (avoid); highly protein bound (displaces phenytoin).

✎ **CORRECTED** Original: valproate is "Category D unless used with folic acid". Folic acid lowers but **does not eliminate** neural-tube risk — valproate remains contraindicated/avoided in pregnancy for epilepsy unless no alternative (and is contraindicated for migraine). L-carnitine is used for **valproate-induced hyperammonaemia / toxicity**.

2 Pregabalin & Gabapentin — α2δ Calcium-Channel Ligands



- **Uses**: **neuropathic pain** (diabetic neuropathy, post-herpetic neuralgia), fibromyalgia (pregabalin), focal seizures (adjunct), generalised anxiety (pregabalin, EU)
- **SE**: dizziness, somnolence, ataxia, **peripheral oedema**, weight gain, blurred vision; **respiratory depression with opioids**
- **Renal elimination**, not metabolised → dose by CrCl; few interactions
- **Abuse/misuse potential**: pregabalin is a **controlled / semi-controlled medicine in the UAE & GCC** (Schedule V in the US); taper to stop

✎ **CORRECTED** Original described pregabalin as a "controlled antipsychotic drug". It is an **antiseizure / analgesic (gabapentinoid)**, not an antipsychotic.

3 Choosing a Neuropathic-Pain Agent

First-line options	+ Hypertension / cardiac disease	+ Depression	+ Insomnia (no cardiac risk)	Trigeminal neuralgia
Pregabalin / gabapentin · duloxetine · TCA (amitriptyline)	**Pregabalin or gabapentin** (SNRIs may ↑ BP; TCAs cardiotoxic)	**Duloxetine** (treats both)	Amitriptyline at night	**Carbamazepine** first-line

Diabetic neuropathy: pregabalin, duloxetine or amitriptyline are all first-line (NICE CG173; ADA). Selection is driven by comorbidities and side effects.

High-Yield Exam Pearls

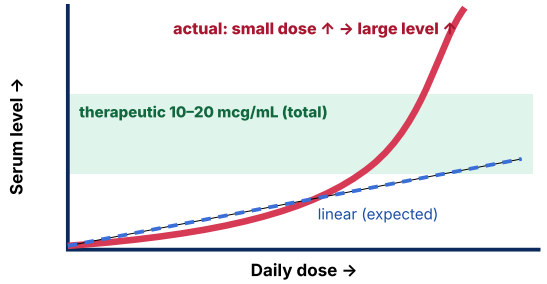
- Valproate: **liver, pancreas, spina bifida**; ammonia → **L-carnitine**.
- Valproate + meropenem → **seizure breakthrough**.
- Valproate + lamotrigine → **halve lamotrigine**.
- Pregabalin: renal dosing, oedema, misuse potential.

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Phenytoin & Fosphenytoin

Pearson Review Notes

1 Non-Linear (Zero-Order) Kinetics



- Hepatic enzymes (CYP2C9/2C19) **saturate** at therapeutic doses → change doses in small steps; recheck levels.
- Usual adult maintenance ≈ 300–400 mg/day, individualised by levels.

✎ **CORRECTED** "Maximum daily dose 500 mg" is not a fixed rule — dosing is individualised by serum level and response.

2 Protein Binding & Albumin Correction

- ~90% bound to **albumin**; only **free** drug is active and crosses the BBB
- Low albumin (malnutrition, liver disease, elderly, pregnancy) → measured total level **underestimates** free drug

SHEINER–TOZER EQUATION

$$C_{corrected} = C_{observed} \div [(0.2 \times albumin) + 0.1]$$

albumin in g/dL · use 0.1 × albumin + 0.1 if CrCl < 10 mL/min

Example: level 8 mcg/mL, albumin 2.0 g/dL → 8 ÷ 0.5 = **16 mcg/mL** (therapeutic — no dose increase needed).

✎ **CORRECTED** The formula corrects the **serum concentration**, not the dose.

3 Formulations

- **Phenytoin sodium** (capsules, IV) vs **phenytoin acid/base** (chewable tablets, suspension): **100 mg sodium ≈ 92 mg phenytoin**
- IV phenytoin: max **50 mg/min** (25 mg/min elderly/cardiac) — propylene glycol → **hypotension, arrhythmias**; in saline only; cardiac monitoring
- **Fosphenytoin**: water-soluble **prodrug**; dosed in **PE** (phenytoin equivalents); can be given **IM**; up to 150 mg PE/min; less infusion-site injury
- Enteral feeds ↓ absorption → hold feeds 1–2 h before/after

4 Adverse Effects

Concentration-related

- **Nystagmus** (> 20 mcg/mL)
- **Ataxia**, diplopia (> 30)
- Lethargy, confusion (> 40)
- Paradoxical seizures at very high levels

Hypersensitivity

- Rash, **SJS/TEN** — **HLA-B*1502** (Asian ancestry); DRESS; hepatitis

Chronic

- **Gingival hyperplasia** (oral hygiene)
- **Hirsutism**, acne, coarse facial features
- **Megaloblastic anaemia** (folate ↓)
- Osteomalacia (vitamin D ↓), peripheral neuropathy
- **Purple glove syndrome** (IV extravasation)
- Pink/red-brown urine (harmless)

5 Uses, Pregnancy & Interactions

- Focal & tonic-clonic seizures, status epilepticus — **not absence** (may worsen) or myoclonic
- **Fetal hydantoin syndrome**: cleft lip/palate, craniofacial & digit defects, growth delay; neonatal **vitamin K**-deficiency bleeding
- **Potent CYP inducer**: ↓ OCs, warfarin (variable), DOACs, many ASMs
- Displaced from albumin by **valproate**, aspirin/high-dose **NSAIDs**, sulfonamides → ↑ free phenytoin

Q. How does phenytoin reach the brain?

a) Bound drug moves from blood by active transport
b) Bound drug moves from blood by diffusion
c) Unbound drug crosses the BBB by simple diffusion
d) Unbound drug crosses the BBB (mechanism unspecified)

Answer: C. Phenytoin is albumin-bound in blood; only the **free (unbound)**, lipophilic fraction crosses the BBB by **passive diffusion**.

High-Yield Exam Pearls

- Low albumin → **correct the level** before changing the dose.
- IV phenytoin ≤ **50 mg/min**; fosphenytoin ≤ 150 mg PE/min.
- Gingival hyperplasia, hirsutism, nystagmus = phenytoin.
- Screen **HLA-B*1502** in Asian patients.

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Carbamazepine, Oxcarbazepine & Lamotrigine

Pearson Review Notes

1 Carbamazepine

Uses

- **First choice for focal seizures** (with lamotrigine)
- Generalised tonic-clonic (not absence/myoclonic)
- **Trigeminal neuralgia — first-line** (sudden, unilateral facial pain)
- Bipolar disorder (ER capsules), neuropathic pain

Kinetics

- Na^+ channel blocker
- **Auto-inducer**: induces its own metabolism over 3–5 weeks → titrate dose
- **Strong CYP3A4 inducer** (↓ OCs, warfarin, many drugs)
- CYP3A4 substrate: ↑ by macrolides, azoles, **grapefruit**
- Level ≈ **4–12 mcg/mL**

Adverse effects

- **Boxed**: **SJS/TEN** (HLA-B*1502 — screen Asians), **aplastic anaemia / agranulocytosis**
- Leukopenia, **thrombocytopenia** (CBC)
- **Hyponatraemia / SIADH**
- Hepatotoxicity, diplopia, dizziness, ataxia
- GI upset → take **with food**
- Teratogen (neural-tube defects)

2 Oxcarbazepine vs Carbamazepine

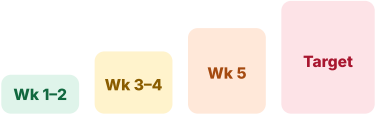
Feature	Carbamazepine	Oxcarbazepine
Chemistry	Parent drug + active epoxide metabolite	**Prodrug** → active MHD (licarbazepine); not a prodrug of carbamazepine
Enzyme induction	Strong (auto-induction)	Weaker (still ↓ OC efficacy); fewer interactions
Hyponatraemia	Yes	**More frequent** — monitor Na^+ (elderly, diuretics)
Blood dyscrasias / hepatic	Aplastic anaemia, agranulocytosis	Rare; no routine CBC needed
Rash	SJS/TEN (HLA-B*1502)	~25–30% cross-sensitivity with carbamazepine

✎ **CORRECTED** Original grouped oxcarbazepine as a "long-acting" prodrug of carbamazepine. It is a **separate prodrug** (→ MHD) with fewer interactions; its signature adverse effect is **hyponatraemia**.

3 Lamotrigine

- Na^+ channel blocker (↓ glutamate release); broad spectrum: focal, GTC, **absence**, Lennox–Gastaut
- **Bipolar maintenance** — prevents depressive episodes
- **Preferred ASM in pregnancy** (with levetiracetam): lowest malformation risk; **levels fall in pregnancy** (↑ glucuronidation) → monitor & adjust; reduce after delivery
- Metabolised by **glucuronidation** (UGT): **estrogen-containing OCs ↓ levels** (~50%); **valproate ↑ levels** (halve dose); inducers (carbamazepine, phenytoin) ↓ levels
- Other SE: dizziness, diplopia, headache, insomnia; rare HLH, aseptic meningitis

Boxed warning — serious rash (SJS/TEN)
Risk ↑ with: rapid titration, exceeding starting dose, **valproate**, children. Use the **slow titration kit**; stop at first sign of rash. If ≥ 5 half-lives (≈ 5 days) missed → **re-titrate from the start**.



Step-wise titration over ≥ 5–6 weeks (doses depend on co-medication — follow the kit/label).

High-Yield Exam Pearls

- Trigeminal neuralgia → **carbamazepine**.
- Carbamazepine: **auto-induction**, SIADH, aplastic anaemia, HLA-B*1502.
- Oxcarbazepine → **hyponatraemia**.
- Lamotrigine + valproate → **halve dose, ↑ SJS risk**.

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Newer ASMs & Special Populations

Pearson Review Notes

1 Other Antiseizure Medicines

Drug	Mechanism	Key points / adverse effects
Levetiracetam	Binds synaptic vesicle protein **SV2A**	Broad spectrum; IV & oral; max **3000 mg/day**; **renal** excretion → adjust when CrCl ≤ 80; not hepatically metabolised → **minimal interactions**; preferred in **pregnancy & breastfeeding**. SE: somnolence, **irritability, aggression, depression, suicidality**
Topiramate	Na^+ block, ↑ GABA-A, AMPA/kainate antagonism, **carbonic anhydrase inhibition**	Focal/GTC, **migraine prophylaxis**, weight loss (with phentermine). Max ≈ 400 mg/day; renal adjust CrCl < 70. SE: cognitive slowing/word-finding, paresthesia, **weight loss**, **kidney stones**, **metabolic acidosis**, **oligohidrosis**/hyperthermia (children), **acute angle-closure glaucoma**; teratogen (cleft lip); ↓ OC efficacy > 200 mg/day
Ethosuximide	**T-type Ca^{2+}** block (thalamus)	**First-line absence** seizures only. SE: GI upset, drowsiness, headache, blood dyscrasias, SJS
Zonisamide	Na^+ + T-type Ca^{2+}; sulfonamide; CA inhibition	**Oligohidrosis**, kidney stones, metabolic acidosis, weight loss; **avoid in sulfonamide allergy**
Lacosamide	Enhances **slow inactivation** of Na^+ channels	Focal seizures; IV/oral. **PR-interval prolongation**, AV block (baseline ECG in cardiac disease); dizziness, diplopia
Vigabatrin	Irreversible **GABA-transaminase** inhibitor	**Infantile spasms**, refractory focal seizures. **Boxed: permanent bilateral visual-field loss** → vision testing
Gabapentin	α2δ Ca^{2+} ligand	Focal (adjunct), post-herpetic neuralgia; renal dosing; sedation, oedema; may worsen myoclonus

✎ **CORRECTED** Original: levetiracetam = "Ca^{2+}-channel blocker and GABA inducer"; topiramate = "Na^+ and Ca^{2+} blocker" only. Levetiracetam acts on **SV2A**; topiramate has **multiple** mechanisms. Topiramate + metformin is a **caution** (additive acidosis), not an absolute contraindication.

2 Drug Selection by Seizure Type

NICE NG217

Seizure type	First-line	Avoid / may worsen
Focal	**Lamotrigine**, **levetiracetam**, carbamazepine, oxcarbazepine	—
Generalised tonic-clonic	**Valproate** (not in women of child-bearing potential), lamotrigine, levetiracetam	—
Absence	**Ethosuximide**, valproate (lamotrigine)	Carbamazepine, phenytoin, gabapentin
Myoclonic	Valproate, levetiracetam	Carbamazepine, lamotrigine (may worsen)
Infantile spasms	ACTH / prednisolone, vigabatrin	—

3 Pregnancy & Contraception

- **Safest**: **lamotrigine, levetiracetam** (monitor lamotrigine levels)
- **Avoid**: **valproate** (highest risk), topiramate, phenytoin, phenobarbital, carbamazepine
- Folic acid **before conception** (higher dose with ASMs)
- Enzyme inducers (carbamazepine, phenytoin, phenobarbital, topiramate > 200 mg) → **OC failure**: use IUD or depot
- Newborns exposed to inducers: **vitamin K** at birth

4 ASM Effects on Liver Enzymes

Strong inducers — carbamazepine, phenytoin, phenobarbital, primidone (weaker: oxcarbazepine, topiramate > 200 mg) → ↓ OCs, warfarin, DOACs, other ASMs

Inhibitor — **valproate** (↑ lamotrigine, phenobarbital, free phenytoin). Felbamate also inhibits

Minimal interactions — levetiracetam, gabapentin, pregabalin, lacosamide → useful in polypharmacy, elderly, transplant

High-Yield Exam Pearls

- Topiramate: **stones, acidosis, glaucoma, weight loss**.
- Vigabatrin → **vision loss**; lacosamide → **PR ↑**.
- Levetiracetam → behaviour change; renal dosing.
- Pregnant woman with epilepsy → **lamotrigine / levetiracetam**.

Attention-Deficit Hyperactivity Disorder (ADHD)

Pearson Review Notes

1 Core Symptoms & Treatment by Age

Inattention
distractible, forgetful

Hyperactivity
fidgety, restless

Impulsivity
interrupts, risky acts

Pre-school (4–5 y)
Behavioural therapy first (parent training); methylphenidate if response inadequate

→

≥ 6 years & adults
Medication + behavioural therapy; stimulants are first-line

✎ **CORRECTED** Original: "< 6 years — therapy only, no medication". Behavioural therapy is first-line, but **methylphenidate may be added** in ages 4–5 when therapy alone is insufficient (AAP 2019).

2 CNS Stimulants — First-Line

Methylphenidate

Blocks **DA & NE reuptake** (DAT, NET). IR 2–3×/day; **extended-release once daily** in the morning (OROS — swallow whole). Dexmethylphenidate = active isomer

Amphetamines

Mixed amphetamine salts, dexamfetamine, **lisdexamfetamine** (prodrug, lower misuse potential). ↑ **release** of DA & NE + reuptake inhibition

Adverse effects

- **Insomnia** (give early in day), **↓ appetite, weight loss**, growth suppression → monitor height/weight
- Headache, abdominal pain, nausea
- **Tachycardia, ↑ BP**, palpitations
- Anxiety, tics; rarely **psychosis / hallucinations**; **misuse potential** (Schedule II US)

Contraindications / cautions

- Use of **MAOIs** within 14 days
- Structural heart disease, arrhythmia, uncontrolled hypertension
- Glaucoma, hyperthyroidism, severe anxiety; substance misuse (prefer non-stimulant)
- Baseline: cardiac history, BP, HR, height, weight

✎ **CORRECTED** Stimulants act mainly on **DA & NE** (not 5-HT); typical GI effects are **appetite loss** and abdominal pain rather than diarrhoea/constipation.

3 Non-Stimulants — Second-Line or When Stimulants Unsuitable

Drug	Mechanism	Key points
Atomoxetine (viloxazine similar)	Selective **NE reuptake inhibitor**	Not controlled; onset 2–4 weeks; **CYP2D6 substrate** (↓ dose with fluoxetine/paroxetine); boxed warning: **suicidal ideation**; rare hepatotoxicity; ↑ HR/BP
Guanfacine ER	Selective **α2A agonist**	Useful with **tics**, oppositional symptoms; sedation, hypotension, bradycardia; **taper** (rebound hypertension)
Clonidine ER	α2 agonist	Sedation (helps insomnia), hypotension; taper to stop

4 Monitoring Checklist

Baseline

Cardiac history & family history of sudden death, BP, HR, height, weight; ECG if cardiac concerns

Every visit

BP, HR, appetite, weight; height in children (growth chart); sleep

Watch for

Mood change, tics, psychosis, suicidality (atomoxetine), misuse/diversion

High-Yield Exam Pearls

- Stimulant + MAOI → **hypertensive crisis**.
- Monitor **growth, BP, HR** on stimulants.
- ADHD + tics / substance misuse → **non-stimulant**.
- Atomoxetine = **NRI**, CYP2D6, suicidality warning.

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Insomnia & Anxiolytics

Pearson Review Notes

1 Insomnia — Treatment Ladder

1. Non-drug
CBT-I (first-line), sleep hygiene, treat causes (pain, depression, caffeine, drugs)

→

2. Short-term drug
Lowest dose, shortest time (≤ 2–4 weeks): Z-drug, orexin antagonist, low-dose doxepin, melatonin (elderly)

→

3. Elderly
Avoid BZDs, Z-drugs, diphenhydramine (Beers: falls, confusion) → melatonin/ramelteon, low-dose doxepin

2 Hypnotic Drugs

Drug / class	Mechanism	Key points
Melatonin	MT1/MT2 agonist (hormone)	Sleep-onset insomnia in elderly (≥ 55 y, PR melatonin), **jet lag**, circadian disorders; well tolerated
Ramelteon	Selective **MT1/MT2 agonist**	**Sleep onset**; **not controlled**, no dependence; improves circadian rhythm. SE: drowsiness, dizziness. **CI with fluvoxamine** (CYP1A2)
Zolpidem	Z-drug: GABA-A (α1/BZ1) modulator	Sleep onset (IR) & maintenance (ER); **women: 5 mg IR** (6.25 mg ER); men 5–10 mg — next-morning impairment
Zaleplon	Z-drug	**Shortest half-life** (~1 h): sleep onset; can be taken in the night if ≥ 4 h remain
Eszopiclone	Z-drug	Onset & maintenance; **no limit on duration** in label; metallic/**bitter taste**
Suvorexant, lemborexant, daridorexant	Dual **orexin receptor antagonists**	Onset & maintenance; somnolence; **CI in narcolepsy**; controlled (C-IV US)
Doxepin (low dose 3–6 mg)	Selective H1 blockade	**Sleep maintenance**; minimal anticholinergic effect at this dose
BZDs · barbiturates	GABA-A ↑ frequency · ↑ duration	BZDs (temazepam, triazolam): short-term only — tolerance, dependence, rebound insomnia. Barbiturates: **not recommended** (low TI, dependence, enzyme induction)

Z-drug boxed warning: complex sleep behaviours (sleep-walking, sleep-driving) — stop if they occur. Z-drugs have **lower but not zero** risk of tolerance, dependence and respiratory depression vs BZDs. In the US (Schedule IV), prescriptions may be refilled up to **5 times within 6 months** — local GCC rules apply.

Herbals: valerian — limited evidence; **kava-kava — hepatotoxic** (restricted in many countries).

3 Buspirone — Anxiolytic, Not a Hypnotic

- **5-HT1A partial agonist**; **generalised anxiety disorder** (second-line / adjunct)
- **Slow onset (2–4 weeks)** — not for acute anxiety or PRN
- **No sedation, no abuse, dependence or withdrawal**; no cross-tolerance with BZDs
- SE: dizziness, nausea, headache; **serotonin syndrome / hypertension with MAOIs** (avoid within 14 days)

✎ **CORRECTED** Original listed buspirone for **insomnia** and listed "abuse, dependence, withdrawal" as effects. Buspirone does **not** treat insomnia and has **no** abuse or dependence potential.

4 Anxiety Treatment Lines

1st line
SSRIs / SNRIs (escitalopram, sertraline, venlafaxine, duloxetine) + CBT

2nd line / adjunct
Buspirone, pregabalin, hydroxyzine

Short-term bridge only
Benzodiazepines (2–4 weeks) while SSRI takes effect

High-Yield Exam Pearls

- Jet lag → **melatonin**; non-controlled hypnotic → **ramelteon**.
- Zolpidem in women → **5 mg**.
- Shortest-acting Z-drug → **zaleplon**.
- GAD without sedation/abuse → **buspirone** (slow onset).

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CNS Rapid Review: Antidotes, Levels & Traps

Pearson Review Notes

1 Toxicity → Antidote / Treatment

Toxicity	Antidote / key treatment
TCA overdose (wide QRS)	**Sodium bicarbonate** IV
Benzodiazepine	**Flumazenil** (caution seizures)
Opioid (with BZD)	**Naloxone**
Lithium	**No antidote** — IV saline, haemodialysis
Serotonin syndrome	Stop drugs, BZD, **cyproheptadine**
NMS	Stop drug, cooling, **dantrolene**, bromocriptine
MAOI hypertensive crisis	**Phentolamine**
Valproate (↑ ammonia)	**L-carnitine**
Acute dystonia	**Benztropine / diphenhydramine**
Anticholinergic delirium	Physostigmine (selected cases)

2 Therapeutic Drug Monitoring

Drug	Usual range	Note
Lithium	0.6–1.2 mEq/L	12-h trough; toxic > 1.5
Phenytoin	10–20 mcg/mL (free 1–2)	Correct for albumin
Carbamazepine	4–12 mcg/mL	Auto-induction
Valproate	50–100 mcg/mL	LFTs, platelets
Clozapine	ANC (label schedule)	Weekly × 6 months

Genetic / lab screens

- **HLA-B*1502** before carbamazepine/phenytoin (Asian ancestry)
- **HLA-A*3101** — carbamazepine hypersensitivity (European, Japanese)
- CYP2C19 poor metabolisers → citalopram max 20 mg
- CYP2D6 — atomoxetine, TCAs, paroxetine/fluoxetine interactions

3 Classic "Drug + Condition" Traps

Bupropion
× epilepsy, bulimia/anorexia

Oxybutynin
× Alzheimer's on ChEI

Metoclopramide
× Parkinson's (use domperidone)

Haloperidol
× Parkinson's / Lewy body

Valproate
× pregnancy, child-bearing age, liver disease

Paroxetine
× pregnancy

TCAs
× post-MI, glaucoma, BPH

Lithium
× with HCTZ, NSAIDs, ACEIs

Carbamazepine
× absence seizures, with clozapine

Phenobarbital
× acute porphyria

Apomorphine
× ondansetron

MAOIs
× tyramine, pseudoephedrine, SSRIs, meperidine

4 Most Important Corrections to the Original Notes

- Lithium has **no antidote** ($NaHCO_3$ is for TCAs)
- Lithium: **avoid first trimester**; not the drug of choice in pregnancy
- Clozapine ANC: **weekly for 6 months**, not every 6 months
- Mirtazapine is an **α2-antagonist**, not a reuptake inhibitor
- Paroxetine: **avoid in pregnancy**; not an epilepsy choice
- Buspirone: anxiety only — **no abuse/dependence**, not for insomnia
- **Never** stop levodopa for on–off; never open the mouth during a seizure
- **Oxybutynin** is wrong for urinary urgency in Alzheimer's
- Gabapentinoids act on **α2δ Ca^{2+}**, not GABA receptors
- Levetiracetam acts on **SV2A**
- Folic acid does **not** make valproate safe in pregnancy
- Status epilepticus first-line: **IV lorazepam / IM midazolam**

High-Yield Exam Pearls

- Read every question for **age, pregnancy, renal/liver function, comorbidity**.
- Ask: "Is there a **serotonergic** or **QT** interaction?"
- Narrow-TI CNS drugs: **lithium, phenytoin, carbamazepine, TCAs**.
- Elderly: avoid **anticholinergics, BZDs, Z-drugs** (Beers).